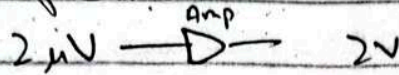
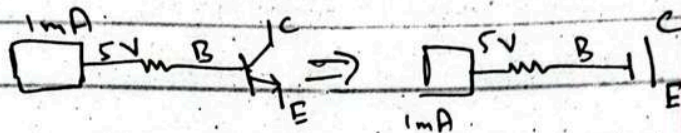


Operational Amplifiers

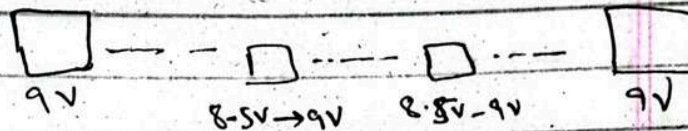
1) Amplification



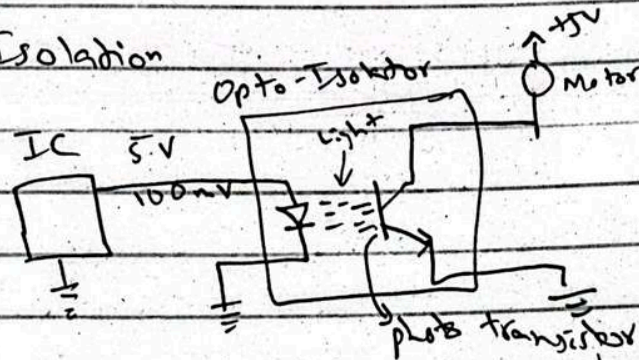
2) Switching



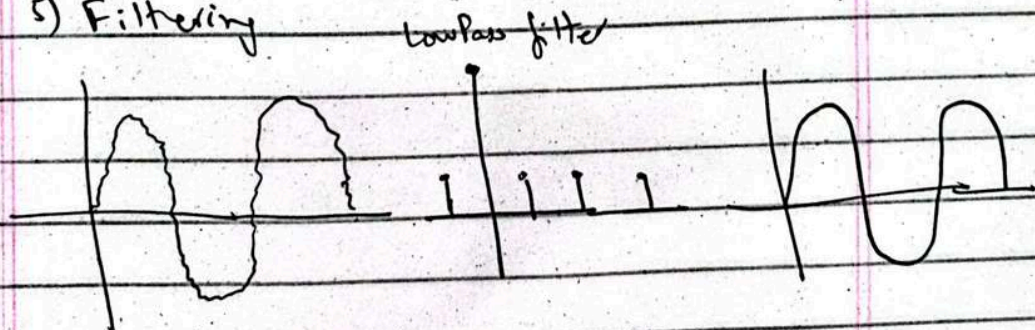
3) Buffering



4) Isolation



5) Filtering



6) Math operations

Signal Amplification:-

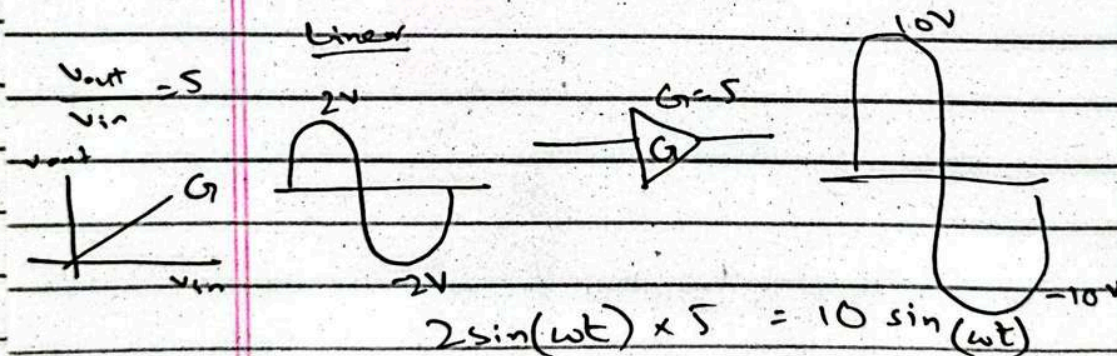
⇒ Most practical sensor generate output of mV, μ V, mA, μ A

⇒ Since the signals are very weak, they need to be amplified



$$V_{out} = G \times V_{in}$$

$$G = \frac{V_{out}}{V_{in}}$$



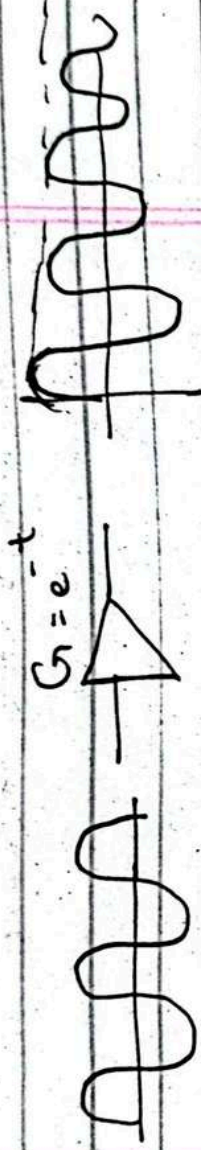
Amplifiers

1) Linear → most preferable amplifier because it doesn't change shape.

2) Non-linear

Linear Amplifier

V_{out} vs V_{in} transfer characteristics is a straight line, hence, linear amplifier or gain G is a constant.



$$V_{out} = V_{in} \times G$$

$$\begin{aligned}\text{Voltage Amplifier} &= \frac{V_{out}}{V_{in}} = A_v \left(\frac{V}{V} \right) \\ \text{Current Amplifier} &= \frac{I_{out}}{I_{in}} = A_i \left(\frac{A}{A} \right) \\ \text{Power Amplifier} &= \frac{P_{out}}{P_{in}} = A_p \left(\frac{W}{W} \right)\end{aligned}$$

Voltage amplifies with op-amp
Current amplifies with BJT

Amplification Gain (dB):

Voltage gain $A_v = \frac{V_o}{V_i} \quad \left(\frac{V}{V}\right)$

$$\Rightarrow A_v(\text{dB}) = 20 \log |A_v|$$

$$= 20 \log \left| \frac{V_o}{V_i} \right| \text{ dB}$$

Current gain $A_i = \frac{I_o}{I_i} \quad \left(\frac{A}{A}\right)$

$$\Rightarrow A_i(\text{dB}) = 20 \log |A_i|$$

$$= 20 \log \left| \frac{I_o}{I_i} \right| \text{ dB}$$

Power gain $A_p = \frac{P_o}{P_i} \quad \left(\frac{W}{W}\right)$

$$P_{\text{max gain}} \Rightarrow 20 \log |A_v|$$

$$= 20 \log \left| \frac{V_o}{V_i} \right|$$

$$= 10 \log \left| \frac{V_o^2}{V_i^2} \right|$$

$$= 10 \log \left| \frac{V_o^2}{V_i^2} \right|$$

$$= 10 \log \left| \frac{V_o^2}{V_i^2} \right|$$

$$P_{A_p}(\text{dB}) = 10 \log \left| \frac{P_o}{P_i} \right| \text{ dB}$$

When the gain is less

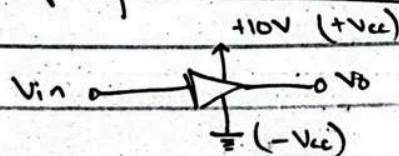
When the value of ~~Any~~ voltage gain is less than 0, it means output is inverted

$$A_v = \frac{V_o}{V_i} < 0 \quad (\text{inverted output})$$

When decibel voltage gain is less than 0, it means output voltage is less than input voltage.

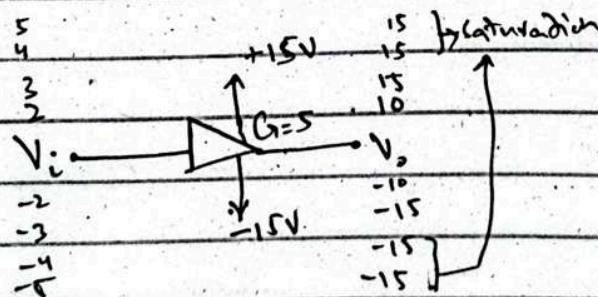
$$A_v (\text{dB}) = 20 \log \left| \frac{V_o}{V_i} \right| < 0 \quad (V_o < V_i)$$

Amplifier Power Supplies & Saturation



Max output of IC $\Rightarrow +V_{cc}$

Min output of IC $\Rightarrow -V_{cc}$



13548000 0.000
135390,00 0

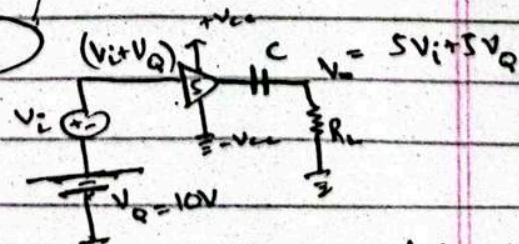
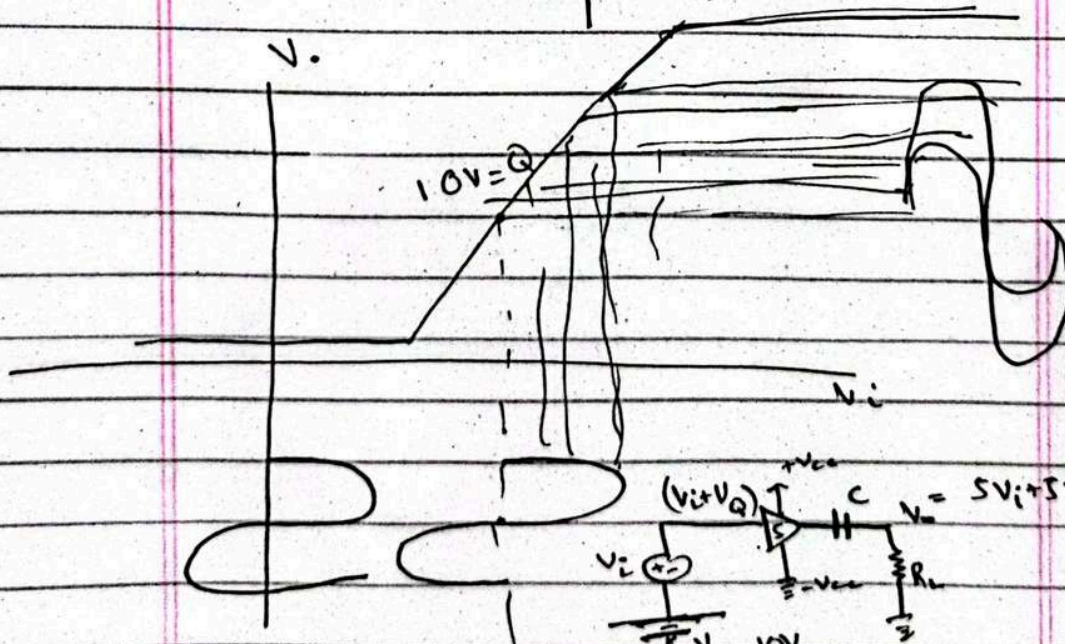
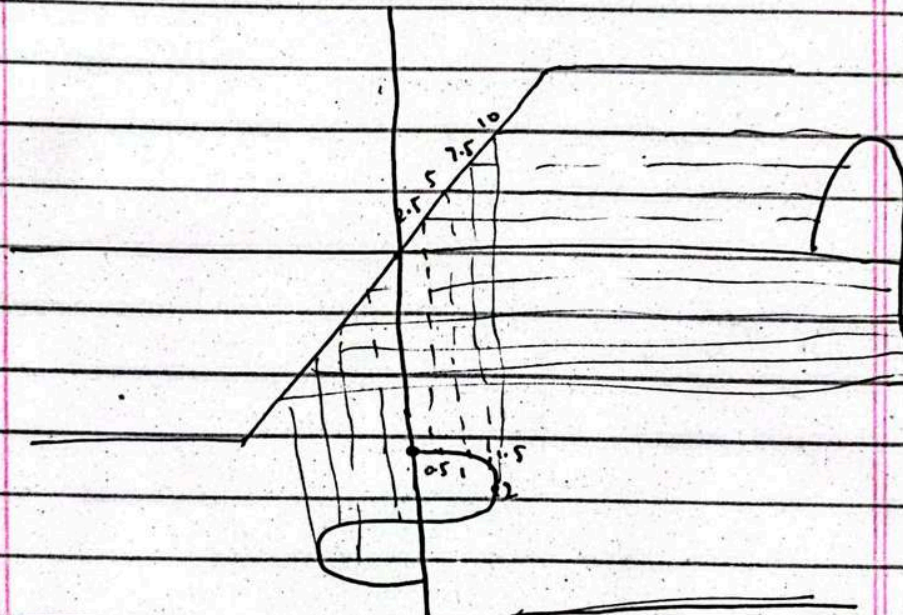
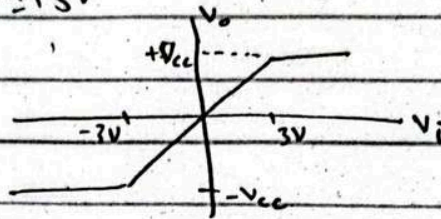
Ideal Transfer Characteristics

$$G = \frac{V_o}{V_i} = T$$

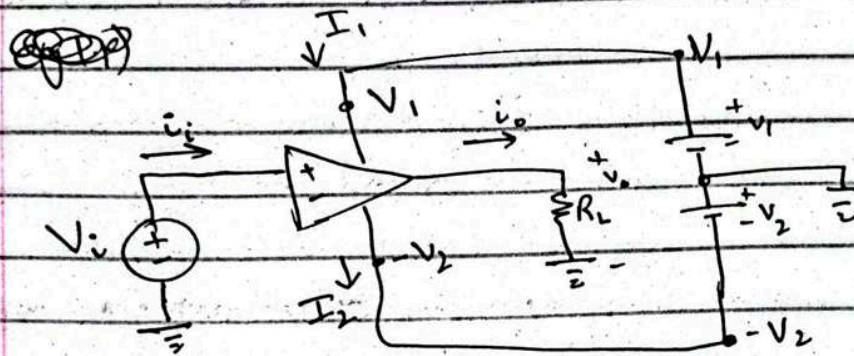
$$V_o = 5V_i$$

$$\Rightarrow y = mx$$

$$+V_{CC} = \pm 15V$$



⇒ If signal is small, then offset is added V_Q then capacitor is used to remove offset (dc input) from output



$$P_i = V_i i_i \quad (\text{input})$$

$$P_o = V_o i_o \quad (\text{output})$$

$$P_{\text{supply}} = V_1 I_1 + V_2 I_2 = 2V_1 I_1 \quad (\text{input})$$

$$i/p = o/p$$

$$P_i + P_{\text{supply}} = P_o + P_{\text{loss}}$$

$$P_{\text{loss}} = P_i + P_{\text{supply}} - P_o$$

$$\eta = \frac{P_o}{P_i + P_{\text{supply}}} \times 100\%$$

Eg 1.1)

$$P_o = \frac{A_o^2}{R_o}$$

$$= \frac{(9)^2}{1k} = 81 \text{ mW}$$

$$V_i \text{ rms} = \frac{1}{\sqrt{2}} \text{ V}$$

$$V_o \text{ rms} = \frac{9}{\sqrt{2}} \text{ V}$$

$$i_i \text{ rms} = \frac{0.1 \text{ m}}{\sqrt{2}} = 0.0707 \text{ mA}$$

$$i_o \text{ rms} = \frac{V_o}{R_L} = \frac{9/\sqrt{2}}{1k} = 6.3 \text{ mA}$$

$$(a) A_v = \frac{V_o}{V_i} = \frac{9}{1} = 9 \quad V/V = 20 \log(9) = 19.08 \text{ dB}$$

$$(b) A_i = \frac{I_o}{I_i} = \frac{6.3 \text{ mA}}{0.0707 \text{ mA}} = 90 \text{ A/A}$$

$$= 20 \log(90) = 39.08 \text{ dB}$$

$$(c) P_o = V_o \text{ rms} \times i_o \text{ rms}$$

$$= \frac{9}{\sqrt{2}} \times 6.3 \text{ mA} = 40.88 \text{ mW}$$

$$P_i = V_i \text{ rms} \times i_i \text{ rms}$$

$$= \frac{1}{\sqrt{2}} \times \frac{0.1}{\sqrt{2}} = 0.05 \text{ mW}$$

$$A_p = \frac{P_o}{P_i} = \frac{40.88 \text{ mW}}{0.05 \text{ mW}} = 810 \text{ W/W}$$

$$A_p = A_v A_i$$

$$= 20 \log(810) = 29.08 \text{ dB}$$

$$P_{\text{supply}} = 2 \times 19 = 38$$

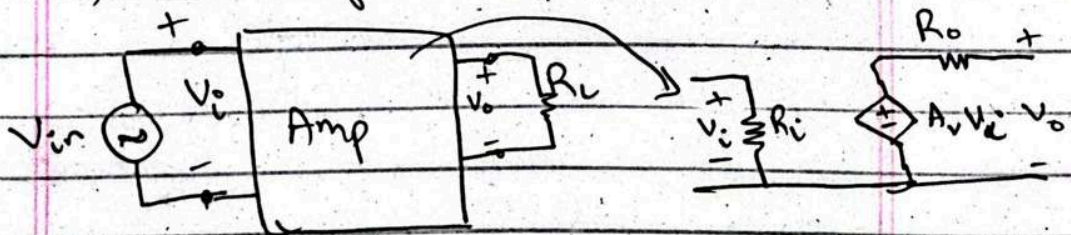
$$P_{\text{supply}} = 2 \times V_i \times I_i = 2 \times 10 \times 9.5 \text{mA} \\ = 190 \text{mW}$$

$$P_{\text{loss}} = P_{\text{supply}} + P_i - P_o \\ = 190 \text{m} + 0.05 \text{m} - 40.5 \text{m} \\ = 149.55 \text{mW}$$

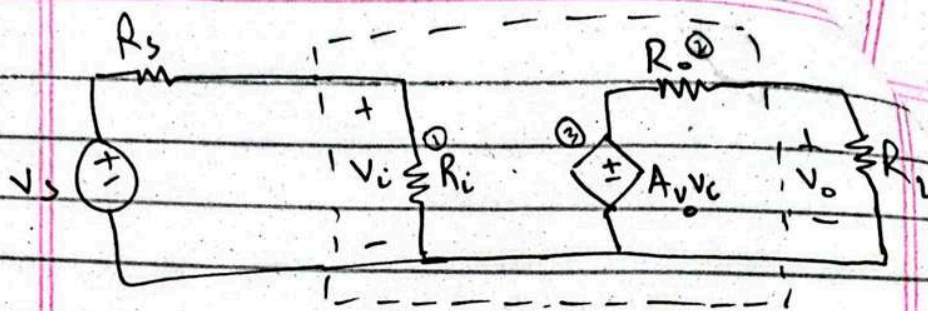
$$\eta = \frac{P_o}{P_i + P_{\text{supply}}} \times 100\% \\ = \frac{40.5 \text{m}}{0.05 \text{m} + 190 \text{m}} \times 100\% \\ = 21.3\%$$

Circuit Model for Amplifiers-

1) Voltage Amplifier



R_i accounts for the fact that amp draws a finite input current from the input voltage source.



$$\rightarrow R_s \ll R_i$$

$$\text{ideally: } R_i \rightarrow \infty$$

$$\rightarrow R_o \ll R_L$$

$$\text{ideally: } R_o \rightarrow 0 \text{ (short circuit)}$$

$$V_i = \left(\frac{R_i}{R_i + R_s} \right) V_s \quad \text{--- (1)}$$

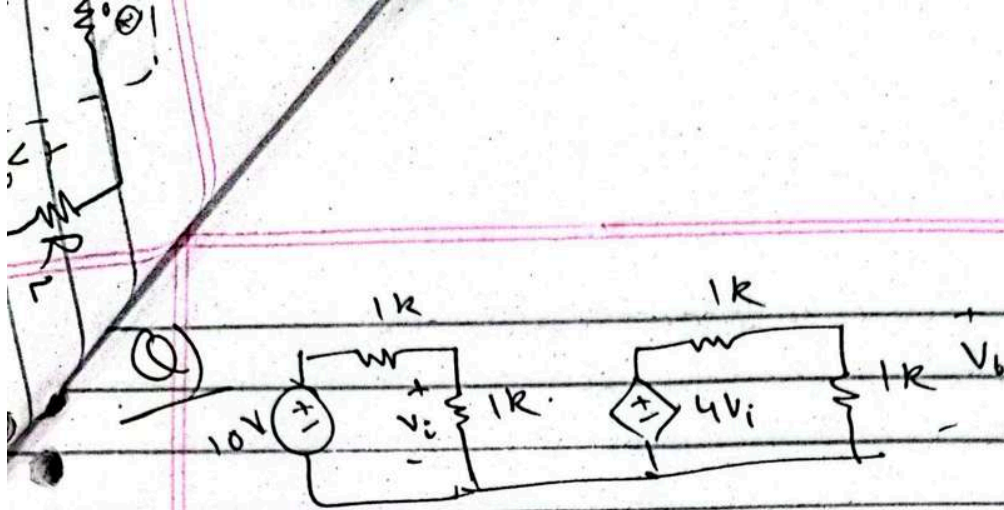
$$V_o = \left(\frac{R_L}{R_L + R_o} \right) A_v V_i \quad \text{--- (2)}$$

(1) in (2)

$$V_o = A_v \left(\frac{R_L}{R_L + R_o} \right) \left(\frac{R_i}{R_i + R_s} \right) V_s$$

$$G = \frac{V_o}{V_s} = A_v \left(\frac{R_L}{R_L + R_o} \right) \left(\frac{R_i}{R_i + R_s} \right)$$

$$\Rightarrow \frac{V_o}{V_i} = A_v \left(\frac{R_L}{R_L + R_o} \right)$$



$$\frac{V_o}{V_s} = A_{v_o} \left(\frac{R_L}{R_L + R_o} \right) \left(\frac{R_i}{R_i + R_s} \right)$$

$$= 4 \left(\frac{1}{2} \right) \left(\frac{1}{2} \right)$$

$$= 1$$

$$V_o = V_s = 10V$$

Buffering

Buffer amplifier
(unity gain)

Current gain:-

$$\Rightarrow I_o = \frac{A_{v_o} V_i}{R_o + R_L}$$

$$\Rightarrow I_i = \frac{V_s}{R_i + R_s}$$

$$\Rightarrow \frac{I_o}{I_i} = \frac{A_{v_o} V_i / R_o + R_L}{V_s / R_i + R_s}$$

$$\frac{I_o}{I_i} = A_{v_o} \left(\frac{V_i}{V_s} \right) \left(\frac{R_i + R_s}{R_o + R_L} \right)$$

$$\Rightarrow V_i = \frac{R_i}{R_s + R_i} V_s$$

$$\frac{V_i}{V_s} = \frac{R_i}{R_s + R_i}$$

$$\begin{aligned} \Rightarrow \frac{i_o}{i_i} &= A_{v_o} \left(\frac{R_i}{R_s + R_i} \right) \left(\frac{R_i + R_s}{R_o + R_L} \right) \\ &= \frac{A_{v_o} R_i}{R_o + R_L} = A_{v_o} \left(\frac{R_i}{R_o + R_L} \right) \end{aligned}$$

OR

$$i_o = \frac{V_o}{R_L}$$

$$i_i = \frac{V_s}{R_s + R_i}$$

$$\begin{aligned} \Rightarrow \frac{i_o}{i_i} &= \frac{V_o / R_L}{V_s / (R_s + R_i)} = \frac{V_o}{V_s} \left(\frac{R_s + R_i}{R_L} \right) \\ &= A_{v_o} \left(\frac{R_i}{R_s + R_i} \right) \left(\frac{R_L}{R_o + R_L} \right) \left(\frac{R_s + R_i}{R_L} \right) \\ &= A_{v_o} \left(\frac{R_i}{R_o + R_L} \right) \end{aligned}$$

Transresistance Gain ($\frac{V}{A}$)

$$\Rightarrow \frac{V_o}{i_i} = \frac{V_o}{V_s/R_s + R_i} = \left(\frac{V_o}{V_s} \right) (R_s + R_i)$$

$$= A_{v_o} \left(\frac{R_i}{R_i + R_s} \right) \left(\frac{R_L}{R_o + R_L} \right) (R_s + R_i)$$

$$= A_{v_o} \left(\frac{R_i R_L}{R_o + R_L} \right)$$

Transconductance ($\frac{A}{V}$)

$$\Rightarrow \frac{i_o}{V_s} = \frac{V_o/R_L}{V_s} = \left(\frac{V_o}{V_s} \right) \left(\frac{1}{R_L} \right)$$

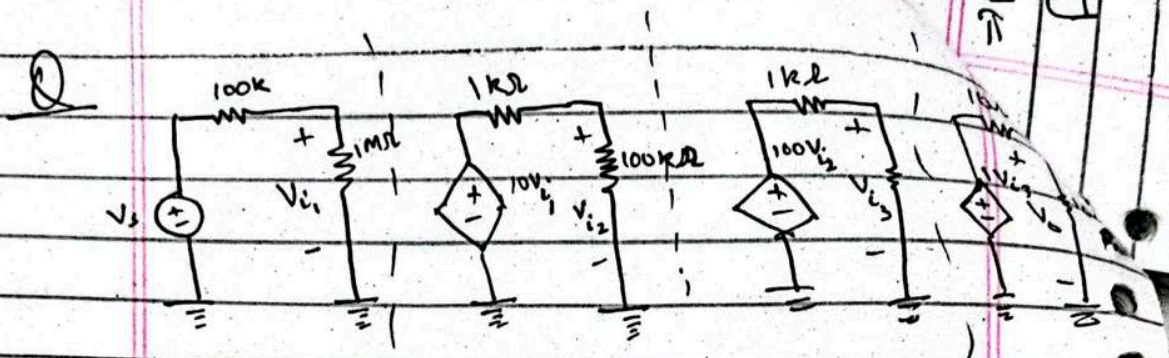
$$= A_{v_o} \left(\frac{R_i}{R_i + R_s} \right) \left(\frac{R_L}{R_o + R_L} \right) \left(\frac{1}{R_L} \right)$$

$$= A_{v_o} \left(\frac{R_i}{(R_i + R_s)(R_o + R_L)} \right)$$

⇒ Cascade Amplifier

Series of amplifiers

$$V_i \rightarrow [G_1] \rightarrow [G_2] \rightarrow [G_3] \rightarrow V_o = (G_1 G_2 G_3) V_i$$



$$\frac{V_o}{V_{i3}} = \frac{100}{100+10} = 0.909 \text{ V/V}$$

$$\frac{V_{i3}}{V_{i2}} = 100 \left(\frac{10k}{1k+10k} \right) = 90.909 \text{ V/V}$$

$$\frac{V_{i2}}{V_{i1}} = 10 \left(\frac{100k}{101k} \right) = 9.900 \text{ V/V}$$

$$\frac{V_{i1}}{V_s} = \frac{1M}{1.1M} = 0.9090 \text{ V/V}$$

$$\begin{aligned} \frac{V_o}{V_s} &= \frac{V_o}{V_{i3}} \times \frac{V_{i3}}{V_{i2}} \times \frac{V_{i2}}{V_{i1}} \times \frac{V_{i1}}{V_s} \\ &= 743.6 \text{ V/V} \\ &= 20 \log(743.6) \text{ dB} \\ &= 57.4 \text{ dB} \end{aligned}$$

$$\begin{aligned} \frac{V_o}{V_{i1}} &= \frac{V_o}{V_{i3}} \times \frac{V_{i3}}{V_{i2}} \times \frac{V_{i2}}{V_{i1}} \\ &= 818.04 \text{ V/V} \\ &= 58.25 \text{ dB} \end{aligned}$$

$$\begin{aligned} \frac{I_o}{I_{i1}} &= \frac{V_o/100}{V_s/100k+1M} \\ &= \frac{V_o}{V_s} (11k) \\ &= (743.6)(11k) = 8.1796 \text{ mA/A} \\ &= 138.25 \text{ dB} \end{aligned}$$

6.08 G

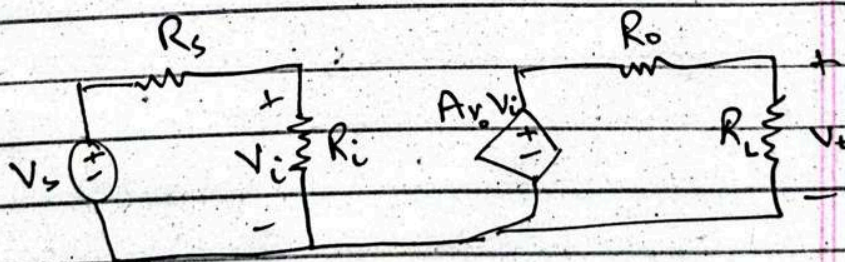
⇒ power gain w.r.t input power

$$\begin{aligned} A_p &= A_v A_i \\ &= \left(\frac{V_o}{V_i} \right) \left(\frac{i_o}{i_i} \right) \\ &= (818.04) (8.18M) \\ &= 6.69 \frac{G \cancel{W}}{\cancel{W}} \\ &= 98.25 \text{ dB} \end{aligned}$$

Four Amplifier Models

- 1) Voltage Amplifiers
- 2) Current Amplifiers
- 3) Trans-Resistance Amplifier ($\frac{V}{A}$)
- 4) Trans-Conductance ($\frac{A}{V}$)

Voltage Amp



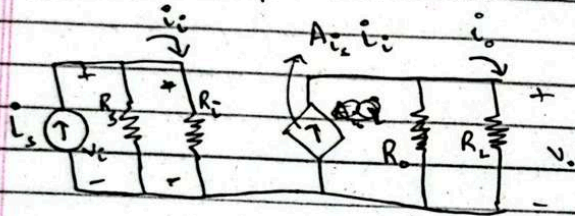
$$i_o = \frac{R_o}{R_o + R_L} A_{i_s} i_i \quad i_i = \frac{R_s}{R_s + R_i} i_s$$

$$V_o = \left(\frac{R_o}{R_o + R_L} \right) A_{i_s} i_i R_L = i_i (R_o \parallel R_L)$$

$$\frac{V_o}{i_s} = \left(\frac{R_o}{R_o + R_L} \right) A_{i_s} \left(\frac{R_s}{R_s + R_i} \right) i_s = \frac{i_i (R_o \parallel R_L)}{i_s}$$

$$\frac{V_o}{i_s} = \frac{i_o R_L}{i_s}$$

Current Amp



$$\Rightarrow i_i = \left(\frac{R_s}{R_s + R_i} \right) i_s$$

$$\Rightarrow i_o = \left(\frac{R_o}{R_o + R_L} \right) A_{i_s} i_i$$

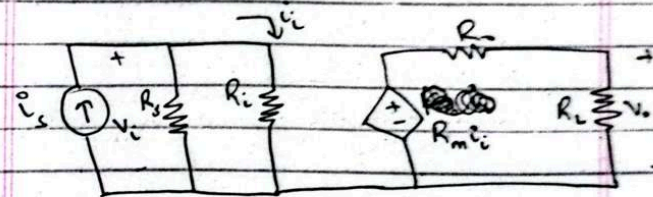
$$\Rightarrow \frac{i_o}{i_s} = A_{i_s} \left(\frac{R_o}{R_o + R_L} \right) \left(\frac{R_s}{R_s + R_i} \right)$$

$$\Rightarrow \frac{V_o}{V_i} = \frac{i_o R_L}{i_s (R_s \parallel R_i)}$$

$$= \left(\frac{i_o}{i_s} \right) \left(\frac{R_L (R_s + R_i)}{R_i R_s} \right)$$

$$\frac{1}{R_s} \parallel \frac{1}{R_i} = \frac{R_i + R_s}{R_i R_s}$$

Trans-Resistance Amplifier $\frac{V_o}{i_s}$

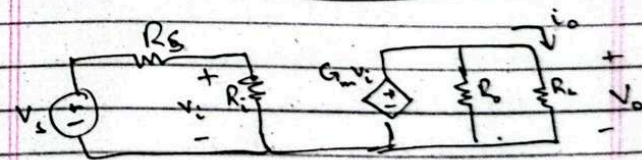


$$\Rightarrow V_o = \left(\frac{R_L}{R_o + R_L} \right) R_m i_i$$

$$\Rightarrow i_i = \left(\frac{R_s}{R_s + R_i} \right) i_s$$

$$\Rightarrow \frac{V_o}{i_s} = R_m \left(\frac{R_L}{R_o + R_L} \right) \left(\frac{R_s}{R_s + R_i} \right)$$

Trans-Conductance Amplifier $\frac{I_o}{V_s}$



$$V_i = \left(\frac{R_i}{R_i + R_s} \right) V_s$$

$$I_o = \left(\frac{R_o}{R_o + R_L} \right) G_m V_i$$

$$\frac{I_o}{V_s} = G_m \left(\frac{R_o}{R_o + R_L} \right) \left(\frac{R_i}{R_i + R_s} \right)$$

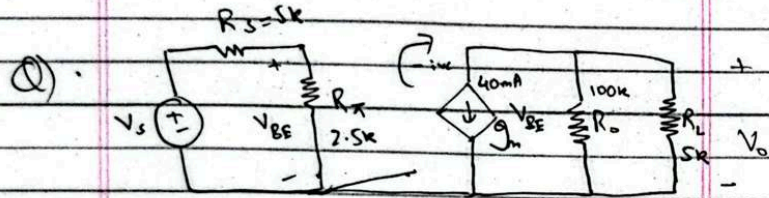
$$\frac{R \parallel R_L}{R \parallel R_L} = \frac{1}{R} + \frac{1}{R_L}$$

$$\frac{R \parallel R_L}{R \parallel R_L} = \frac{1}{R} + \frac{1}{R_L}$$

$$R \parallel R_L = \frac{R \cdot R_L}{R + R_L}$$

$$I_c = \beta I_B$$

BJT Model (Transconductance Amp)



$$V_o = -g_m V_{BE} (R_o \parallel R_L)$$

$$= -g_m V_{BE} \left(\frac{R_o R_L}{R_o + R_L} \right)$$

$$V_{BE} = \frac{R_L}{R_s + R_L} V_s$$

$$V_o = -g_m V_s \left(\frac{R_L}{R_s + R_L} \right) \left(\frac{R_o R_L}{R_o + R_L} \right)$$

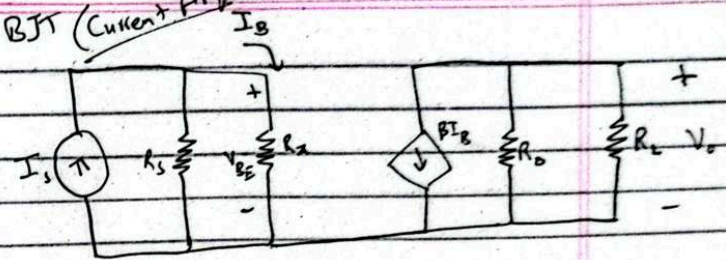
$$\frac{V_o}{V_s} = - (40m) \left(\frac{2.5k}{2.5k + 5k} \right) \left(\frac{(100k)(5k)}{100k + 5k} \right)$$

$$= -63.49 \%$$

$$I_o = \frac{R_o}{R_o + R_L} g_m V_{BE}$$

$$\Rightarrow I_o = \frac{R_o}{R_o + R_L} g_m \frac{R_L}{R_s + R_L} V_s$$

Ex 1.4 BJT (Current Amp)



$$g_m V_{BE} = \beta I_B$$

$$\beta = g_m \frac{V_{BE}}{I_B}$$

$$= g_m R_L$$

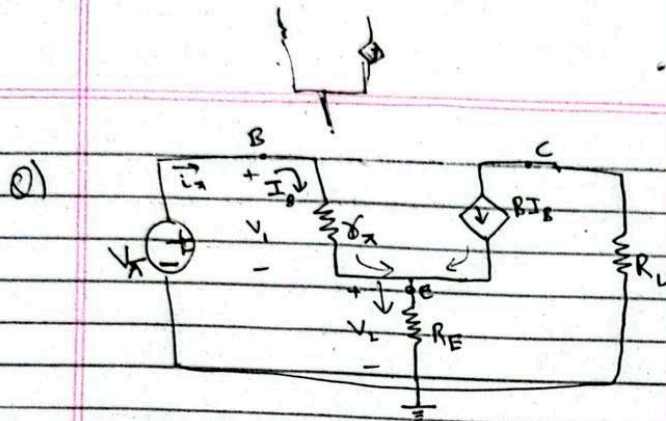
$$= (40m) (2.5k)$$

$$= 100 A$$

$$\frac{I_o}{V_s} = g_m \frac{R_o}{R_s + R_L} \frac{R_L}{R_s + R_L}$$

$$= (-40m) \left(\frac{100k}{100k + 5k} \right) \left(\frac{2.5k}{2.5k + 5k} \right)$$

$$= -0.0127$$



Find R_{internal} of this amplifier

$$\frac{V_x}{i_x} = R_{\text{int}}$$

$$V_x = V_1 + V_2$$

$$= I_B r_x + (I_B + \beta I_B) R_E$$

$$= I_B r_x + \beta (1 + \beta) I_B R_E$$

$$= I_B (r_x + (1 + \beta) R_E)$$

$$\frac{V_x}{I_B} = r_x + (1 + \beta) R_E$$

A) $i_x = I_B$

$$\frac{V_x}{I_B} = r_x + (1 + \beta) R_E$$

$$R_{\text{int}} = r_x + (1 + \beta) R_E$$

Single-Time Constant Circuit: Those circuit having 1 time constant e.g. RL, RC circuit
(reactive element)
(capacitor, inductor)

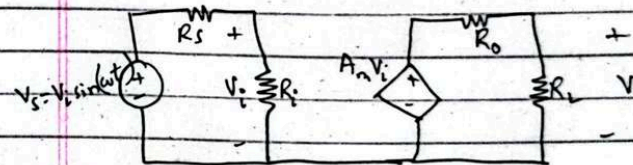
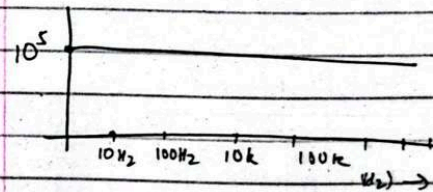
Frequency Response of Amplifiers

Axioms:-

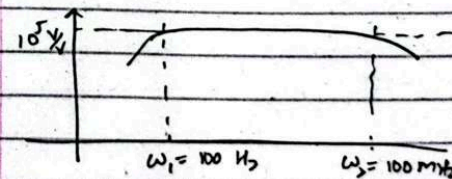
- 1) Frequency remains same
- 2) Shape of waveform remains same
- 3) Amplitude changes
- 4) Phase angle changes

capacitor is placed in series
No output capacitor
remove its effect

Magnitude/Gain of an 'Ideal' voltage Amp



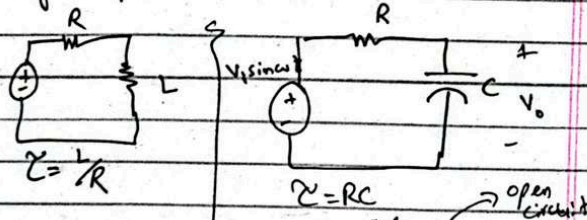
Magnitude/Gain of 'Real' Volt. Amp



capacitor is placed in series
No output capacitor
remove its effect

Single-time Constant Circuit:-

circuits having one time constant
e.g. RL/RC circuits

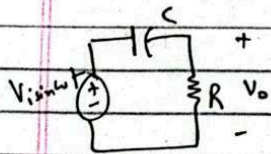
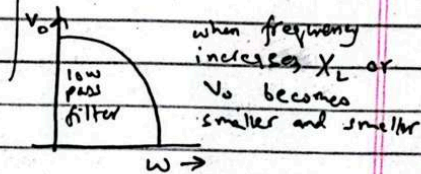


$$\omega \rightarrow 0; X_L \rightarrow \infty$$

$$V_o = V_i$$

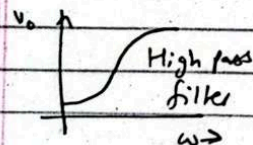
$$\omega \rightarrow \infty; X_C \rightarrow 0$$

$$V_o = 0$$

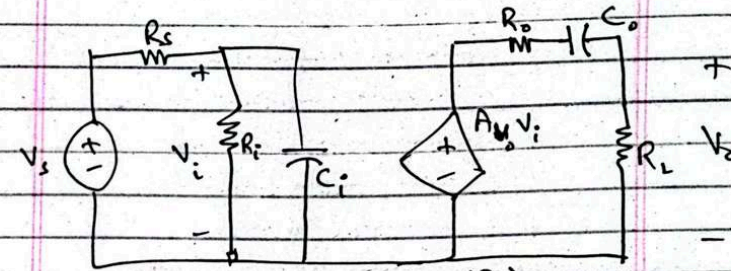


$$\omega \rightarrow 0; X_C \rightarrow \infty; V_o = 0 \text{ (small)}$$

$$\omega \rightarrow \infty; X_C \rightarrow 0; V_o = V_i$$



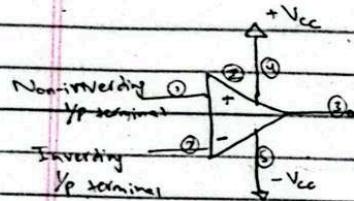
⇒ In RC circuit, when output voltage is at resistor, it is high pass filter & when output voltage is at capacitor, it is low pass filter.



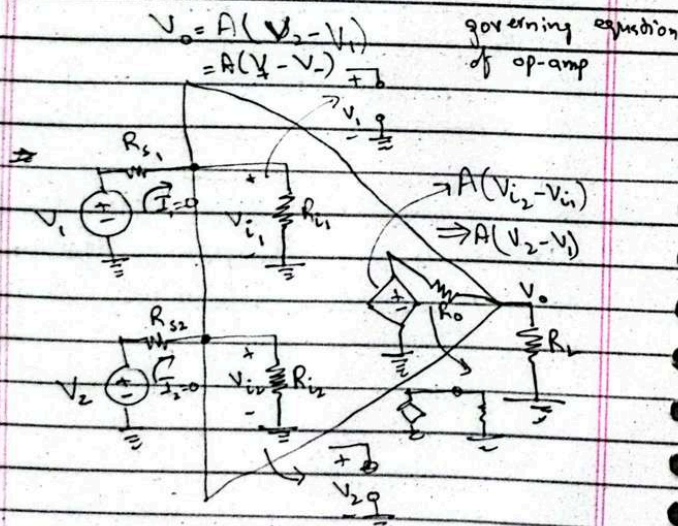
⇒ We place input capacitor parallel to input resistor (R_i) to filter all noise and hold the signal.

⇒ We place output capacitor (C_o) in series with output resistor (R_o) to remove any DC offset (dc input).

Operational Amplifiers



(a) $\Rightarrow$ Operational amplifier is used to amplify the difference b/w inverting and non-inverting inputs V_1 and V_2 .



(b) $\Rightarrow$ Input current from signal V_1 & V_2 is zero.

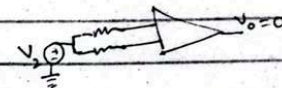
$\Rightarrow$ Op-amp does not draw any current from V_1 & V_2 .

$$R_{i1} \rightarrow \infty$$

$$R_{i2} \rightarrow \infty$$

$$(c) \Rightarrow R_o = 0$$

$\Rightarrow$ If input is same i.e. $V_1 = V_2$, output is zero ideally (common mode rejection).



$$V_0 = A(V_2 - V_1) \Rightarrow A_{cm} = \frac{V_0}{V_2 - V_1}$$

$$A_{cm} = \frac{V_0}{V_i} \text{ (ideally } = 0 \text{)}$$

$\Rightarrow$ Greater the A_{cm} , the bad the op-amp is.

$\Rightarrow$ Common mode rejection ratio

$$CMRR = \left| \frac{A_d}{A_{cm}} \right| \text{ (ideally } = \infty \text{)}$$

$$= 20 \log \left| \frac{A_d}{A_{cm}} \right| \text{ dB}$$

Greater CMRR, better the op-amp

(d) C_i & C_o are DC coupled capacitors

$$(e) CMRR \rightarrow \infty$$

Saturation $\longleftrightarrow$ Attenuation

(f) $A \rightarrow \infty$ (ideally)
 $A = 10^5$ (practically)

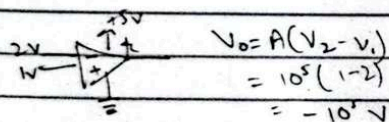
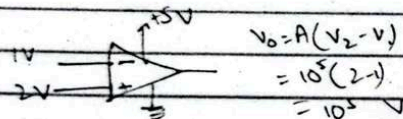
(g) Infinite bandwidth



OP-AMP two way

1) Open-loop

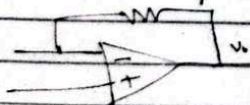
~~2) Closed-loop~~



$\Rightarrow V_+ > V_- \Rightarrow V_o = +V_{cc}$

$\Rightarrow V_- > V_+ \Rightarrow V_o = -V_{cc}$

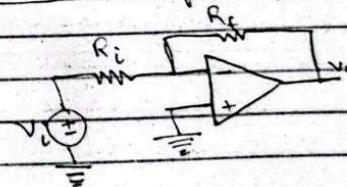
2) Closed Loop



$\Rightarrow$ In open-loop configuration, op-amp can only be used as a comparator. You cannot adjust the gain as per your choice.

$\Rightarrow$ In closed-loop configuration, the design becomes more flexible and you can change the gain as per your choice.

Inverting Amplifier



If loop is closed,
 $V_- = V_+$

Case-1 ($A \rightarrow \infty$)

$$V_o = A(V_+ - V_-)$$

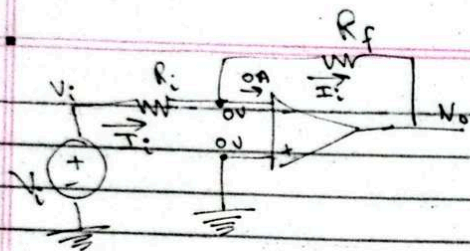
$$V_+ - V_- = \frac{V_o}{A}$$

$$\Rightarrow A \rightarrow \infty ; \frac{V_o}{A} \rightarrow 0$$

$$V_+ - V_- = 0$$

$$V_+ = V_-$$

(Virtual Short circuit)



$$V_+ = 0V$$

$$(V_+ = V_-)$$

$$\Rightarrow V_- = 0V$$

$$\Rightarrow I_i = \frac{V_i - 0}{R_i} = \frac{V_i}{R_i} \quad \text{--- (1)}$$

$$\Rightarrow 0 - V_o = I_i R_f$$

$$V_o = -I_i R_f \quad \text{--- (2)}$$

$$\Rightarrow V_o = -\left(\frac{V_i}{R_i}\right) R_f$$

$$\Rightarrow \frac{V_o}{V_i} = -\frac{R_f}{R_i} = G$$

~~SG~~

Nodal Analysis

$$\frac{0 - V_i}{R_i} + \frac{0 - V_o}{R_f} = 0$$

$$-\frac{V_i}{R_i} - \frac{V_o}{R_f} = 0$$

$$\frac{V_o}{R_f} = -\frac{V_i}{R_i}$$

$$\frac{V_o}{V_i} = -\frac{R_f}{R_i}$$

Case-II ($A \rightarrow \text{Finite}$)

$$V_o = A(V_+ - V_-)$$

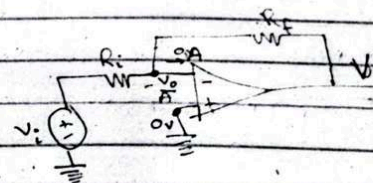
$$V_+ - V_- = \frac{V_o}{A}$$

$$\Rightarrow A \rightarrow \text{finite}$$

$$\& V_- = V_+ - \frac{V_o}{A}$$

$$\Rightarrow V_+ = 0V$$

$$V_- = 0 - \frac{V_o}{A}$$



(Nodal Analysis) $\frac{-\frac{V_o}{A} - V_i}{R_i} + \frac{-\frac{V_o}{A} - V_o}{R_f} = 0$

$$-\frac{V_o}{AR_i} - \frac{V_i}{R_i} - \frac{V_o}{AR_f} - \frac{V_o}{R_f} = 0$$

$$-\frac{V_o}{AR_i} - \frac{V_o}{AR_f} - \frac{V_o}{R_f} = \frac{V_i}{R_i}$$

$$\left(\frac{1}{AR_i} + \frac{1}{AR_f} + \frac{1}{R_f}\right) V_o = -\frac{V_i}{R_i}$$

$$V_o \left(\frac{1}{A \left(\frac{1}{R_i} + \frac{1}{R_f} \right) + \frac{1}{R_f}} \right) = -\frac{V_i}{R_i}$$

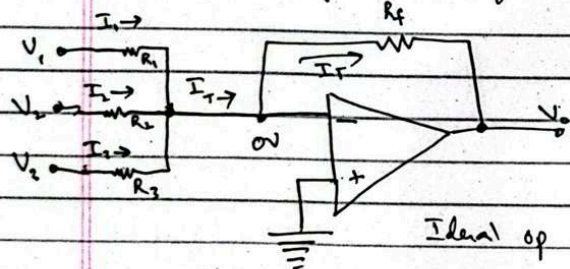
Multiply R_i on b.s

$$V_o \left(\frac{1}{A} \left(\frac{R_F}{R_i} + 1 \right) + 1 \right) = \left(-\frac{R_F}{R_i} \right) V_i$$

$$G = \frac{V_o}{V_i} = \frac{-R_F/R_i}{\frac{1}{A} \left(\frac{R_F}{R_i} + 1 \right) + 1}$$

Weighted Summing Amplifier

⇒ Implementing Addition of Multiple Signals



$$V_o = A(V_+ - V_-)$$

$$\frac{V_o}{A} = V_+ - V_-$$

$$A \rightarrow \infty; \quad 0 = V_+ - V_-$$

$$V_+ = V_-$$

$$I_1 = \frac{V_1}{R_1}, \quad I_2 = \frac{V_2}{R_2}, \quad I_3 = \frac{V_3}{R_3}$$

⇒ For no weightage,

$$R_F = R_1 = R_2 = R_3 = \dots$$

$$\text{KCL } I_1 = I_2 + I_3$$

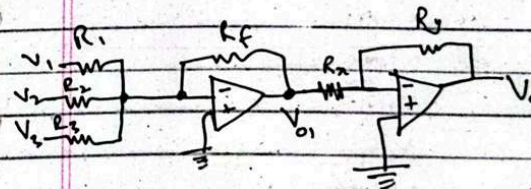
$$= \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_2} = 0$$

$$\Rightarrow 0 - V_o = I_1 R_F$$

$$V_o = -I_1 R_F$$

$$= -R_F \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_2} \right)$$

$$= - \left[\left(\frac{R_F}{R_1} \right) V_1 + \left(\frac{R_F}{R_2} \right) V_2 + \left(\frac{R_F}{R_2} \right) V_3 \right]$$



$$V_{o1} = - \left(\left(\frac{R_F}{R_1} \right) V_1 + \left(\frac{R_F}{R_2} \right) V_2 + \left(\frac{R_F}{R_3} \right) V_3 \right)$$

$$V_o = - \left(\frac{R_3}{R_2} \right) V_{o1}$$

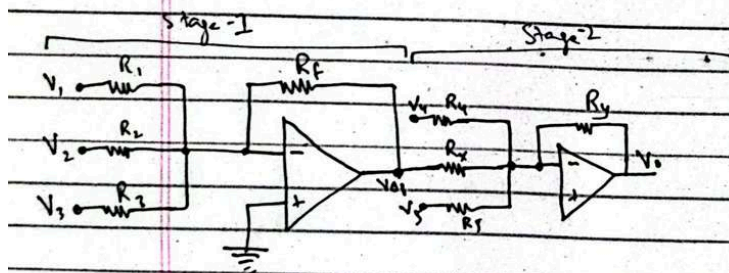
$$= + \left(\frac{R_3}{R_2} \right) \left(\frac{R_F}{R_1} \right) V_1 + \left(\frac{R_3}{R_2} \right) \left(\frac{R_F}{R_2} \right) V_2 + \left(\frac{R_3}{R_2} \right) \left(\frac{R_F}{R_3} \right) V_3$$

For positive gain, $R_3 = R_2$

→ Implementing Addition/Subtraction Circuit

In case of addition and subtraction, we use a 2-stage amplifier design. All those voltages to be added are made inputs of inverting terminal of stage-1 where all those voltages to be subtracted are made inputs of inverting terminal of stage-2 of amp.

$$V_o = \alpha V_1 + \beta V_2 + \gamma V_3 - \epsilon V_4 - \phi V_5$$



$$V_{01} = -\frac{R_F}{R_1} V_1 - \frac{R_F}{R_2} V_2 - \frac{R_F}{R_3} V_3$$

$$V_o = -\frac{R_Y}{R_X} V_{01} - \frac{R_Y}{R_4} V_4 - \frac{R_Y}{R_5} V_5$$

$$= -\frac{R_Y}{R_X} \left[-\frac{R_F}{R_1} V_1 - \frac{R_F}{R_2} V_2 - \frac{R_F}{R_3} V_3 \right] - \frac{R_Y}{R_4} V_4 - \frac{R_Y}{R_5} V_5$$

$$= \left(\frac{R_Y}{R_X} \right) \left(\frac{R_F}{R_1} \right) V_1 + \left(\frac{R_Y}{R_X} \right) \left(\frac{R_F}{R_2} \right) V_2 + \left(\frac{R_Y}{R_X} \right) \left(\frac{R_F}{R_3} \right) V_3 - \frac{R_Y}{R_4} V_4 - \frac{R_Y}{R_5} V_5$$

$$= \alpha V_1 + \beta V_2 + \gamma V_3 - \epsilon V_4 - \phi V_5$$

Q) $V_o = 2V_1 + 8V_2 + 5V_3 - 6V_4 - 4V_5$

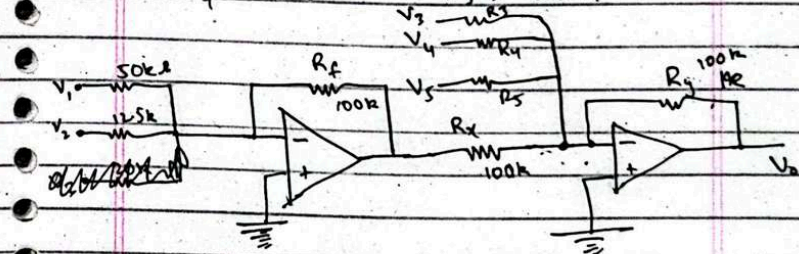
$\Rightarrow R_F = 100k$
 $\Rightarrow R_Y = 100k$

$$\frac{R_F}{R_1} = 2, \quad \frac{R_F}{R_2} = 8, \quad \frac{R_Y}{R_4} = 6$$

$$\frac{R_F}{R_3} = 5, \quad \frac{R_Y}{R_5} = 4$$

$$\Rightarrow R_1 = 50k, \quad R_2 = 12.5k, \quad R_3 = 20k$$

$$R_4 = 16.67k, \quad R_5 = 25k$$

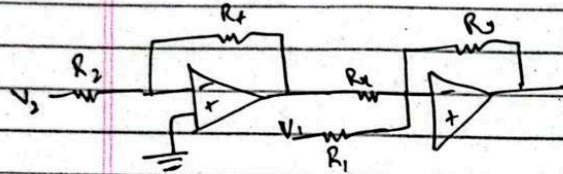


$$V_o = \frac{R_F}{R_1} \frac{R_Y}{R_X} V_1 + \frac{R_F}{R_2} \frac{R_Y}{R_X} V_2 + \frac{R_F}{R_3} \frac{R_Y}{R_X} V_3 - \frac{R_Y}{R_4} V_4 - \frac{R_Y}{R_5} V_5$$

$$\frac{V}{R} = I \quad V = 4R \quad 100 = (4)(25)$$

Difference Amplifier

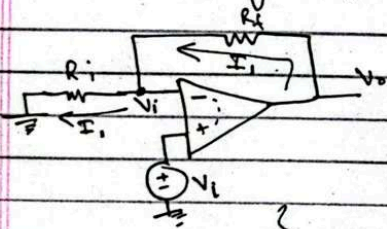
① $V_o = V_2 - V_1$ $R_3 = R_4 = 100k\Omega$



$$\frac{R_4}{R_3} = 1 \Rightarrow R_2 = 100k$$

$$\frac{R_3}{R_1} = 1 \Rightarrow R_1 = 100k$$

Non-Inverting Amp



$A \rightarrow \infty$; $V_o = A(V_+ - V_-)$ $\frac{V_i - 0}{R_i} + \frac{V_i - V_o}{R_f} = 0$

$$\frac{V_o}{A} = V_+ - V_-$$

$$V_+ = V_-$$

$$V_+ = V_i$$

$$V_- = V_+ = V_i$$

$$\frac{V_i}{R_i} + \frac{V_i - V_o}{R_f} = 0$$

$$V_i \left(\frac{1}{R_i} + \frac{1}{R_f} \right) = \frac{V_o}{R_f}$$

$$V_o = \left(\frac{R_f}{R_i} + \frac{R_f}{R_f} \right) V_i$$

$$\frac{V_o}{V_i} = 1 + \frac{R_f}{R_i}$$

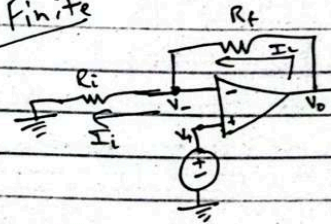
$$I_i = \frac{V_i - 0}{R_i} = \frac{V_i}{R_i}$$

$$\frac{V_o - V_i}{R_f} = I_i$$

$$\frac{V_o}{R_f} - \frac{V_i}{R_f} = I_i \Rightarrow I_i = \frac{V}{R}$$

$$\frac{V_o}{R_f} = \frac{V_i}{R_i} + \frac{V_i}{R_f}$$

$A \rightarrow \text{Finite}$



$$V_o = A(V_+ - V_-)$$

$$\frac{V_o}{A} = V_+ - V_-$$

$$\Rightarrow V_- = V_+ - \frac{V_o}{A} \quad \text{--- (1)}$$

$$\Rightarrow V_+ = V_i$$

$$\Rightarrow V_- = V_i - \frac{V_o}{A}$$

$$\Rightarrow I_i = \frac{(V_i - \frac{V_o}{A}) - 0}{R_i}$$

$$I_i = \frac{V_i}{R_i} - \frac{V_o}{AR_i}$$

$$\Rightarrow V_o - (V_i - \frac{V_o}{A}) = I_i R_F$$

$$V_o - V_i + \frac{V_o}{A} = I_i R_F$$

$$V_o - V_i + \frac{V_o}{A} = \left(\frac{V_i - V_o}{R_i} \right) R_F$$

$$= \left(\frac{R_F}{R_i} \right) V_i - \left(\frac{R_F}{R_i} \right) \left(\frac{V_o}{A} \right)$$

$$V_o + \frac{V_o}{A} + \left(\frac{R_F}{R_i} \right) \left(\frac{V_o}{A} \right) = \frac{R_F}{R_i} V_i + V_i$$

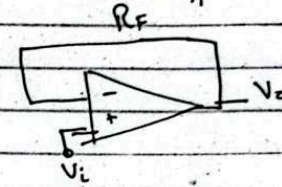
$$V_o \left(1 + \frac{1}{A} \left(1 + \frac{R_F}{R_i} \right) \right) = V_i \left(1 + \frac{R_F}{R_i} \right)$$

$$\frac{V_o}{V_i} = \frac{1 + \frac{R_F}{R_i}}{1 + \frac{1}{A} \left(1 + \frac{R_F}{R_i} \right)}$$

$\Rightarrow$ When we get data from sensor, it is compulsory to pass output sensor data from a voltage follower so data is lost.



Unity Gain Buffer (Voltage Follower)



$$\frac{V_o}{V_i} = 1 + \frac{R_F}{R_i}$$

$$\Rightarrow R_F = 0 \Rightarrow \frac{V_o}{V_i} = 1 \quad (\text{short circuit})$$

$$\Rightarrow R_i = \infty \Rightarrow \frac{V_o}{V_i} = 1 \quad (\text{open circuit})$$

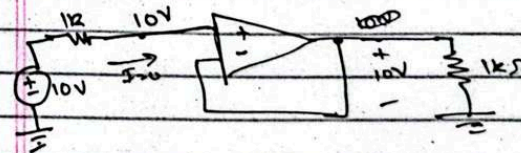
$$V_o = A(V_+ - V_-)$$

$$V_+ - V_- = \frac{V_o}{A}$$

$$A \rightarrow \infty; \text{ ideal}$$

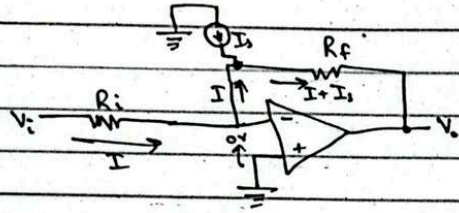
$$V_+ - V_- = 0$$

$$\Rightarrow V_+ = V_i = V_-$$



Special Case of Inv. Amp

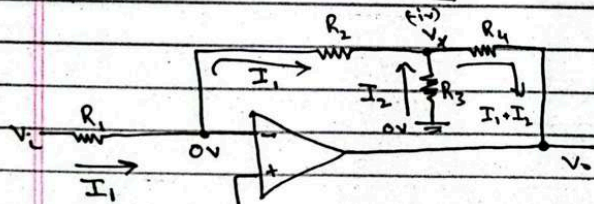
- ⇒ Problem: large gain but also large input resistance
- ⇒ Objective: $R_i \uparrow$ $G \uparrow$ R_f (feasible)



$$0 - V_o = IR_f$$

$$V_o = -IR_f$$

$$V_o = -(I + I_s)R_f$$



$$\Rightarrow I_1 = \frac{V_i}{R_1} \quad \text{--- (1)}$$

$$\Rightarrow 0 - V_x = I_1 R_2$$

$$V_x = -I_1 R_2 \quad \text{--- (2)}$$

$$\Rightarrow 0 - V_x = I_2 R_3$$

$$V_x = -I_2 R_3 \quad \text{--- (3)}$$

$$\Rightarrow V_x - V_o = (I_1 + I_2) R_4 \quad \text{--- (4)}$$

From (1) & (2)

$$V_x = -I_1 R_2 = -\left(\frac{R_2}{R_1}\right) V_i$$

From (3)

$$V_x = -I_2 R_3$$

$$I_2 = \frac{-V_x}{R_3} = \left(\frac{R_2}{R_1 R_3}\right) V_i$$

From (4)

$$V_x - V_o = (I_1 + I_2) R_4$$

$$-\left(\frac{R_2}{R_1}\right) V_i - V_o = \left(\frac{V_i}{R_1} + \frac{R_2 V_i}{R_1 R_3}\right) R_4$$

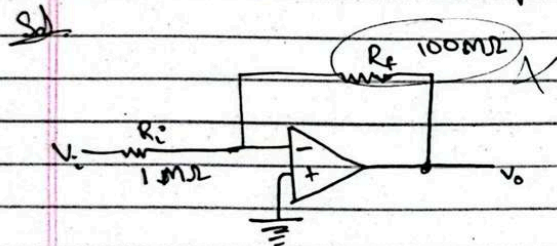
$$-V_o = \frac{R_4}{R_1} V_i + \frac{R_2 R_4}{R_1 R_3} V_i + \frac{R_2}{R_1} V_i$$

$$V_o = -\left(\frac{R_4}{R_1} + \frac{R_2 R_4}{R_1 R_3} + \frac{R_2}{R_1}\right) V_i$$

$$\frac{V_o}{V_i} = -\left(\frac{R_2}{R_1} + \frac{R_4}{R_1} + \frac{R_2 R_4}{R_1 R_3}\right)$$

$$0 - V_x = I_1 R_{10K}$$

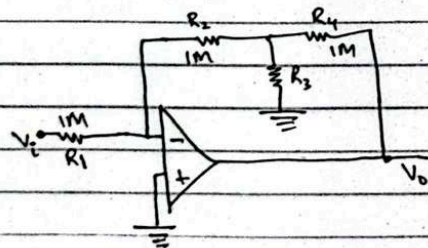
d) Design a normal inv. amp. of gain -100. $R_i = 1\text{M}\Omega$. To increase practicality, design an improved ~~amp~~ T-network inverting amplifier. All resistances should be less than or equal to $1\text{M}\Omega$.



Not practically possible.

$$R_i = 1\text{M}\Omega$$

$$\text{Let, } R_2 = R_4 = R_1 = 1\text{M}\Omega$$



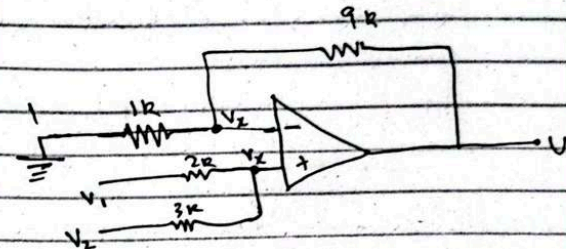
$$\frac{V_o}{V_i} = - \left(\frac{R_2}{R_i} + \frac{R_4}{R_i} + \frac{R_2 R_4}{R_i R_3} \right)$$

$$-100 = - \left(\frac{1\text{M}}{1\text{M}} + \frac{1\text{M}}{1\text{M}} + \frac{(1\text{M})(1\text{M})}{(1\text{M}) R_3} \right)$$

$$100 - 2 = \frac{1\text{M}}{R_3}$$

$$R_3 = \frac{1\text{M}}{98} = 10.204\text{k}\Omega$$

Q)



$$V_x = V_- = V_+$$

Non-Inv. Terminal

$$\frac{V_x - V_1}{2\text{k}} + \frac{V_x - V_2}{3\text{k}} = 0$$

$$3V_x - 3V_1 + 2V_x - 2V_2 = 0$$

$$5V_x = 3V_1 + 2V_2 \quad \text{--- (1)}$$

Inv. Terminal

$$\frac{V_x - 0}{1\text{k}} + \frac{V_x - V_o}{9\text{k}} = 0$$

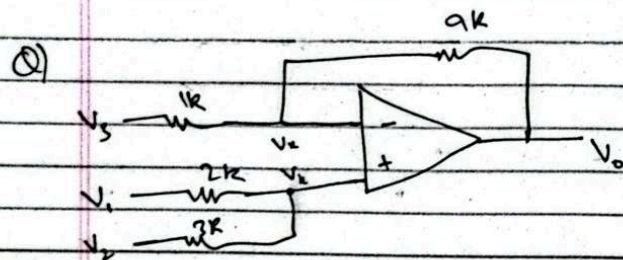
$$9V_x + V_x - V_o = 0$$

$$V_o = 10V_x \quad \text{--- (2)}$$

$$V_o = 2(5V_x)$$

$$= 2(3V_1 + 2V_2)$$

$$V_o = 6V_1 + 4V_2$$



$$-V_+ = V_- = V_x$$

$$\Rightarrow \frac{V_x - V_1}{2k} + \frac{V_x - V_2}{3k} = 0$$

$$5V_x = 3V_1 + 2V_2$$

$$\Rightarrow \frac{V_x - V_3}{1k} + \frac{V_x - V_o}{9k} = 0$$

$$9V_x - 9V_3 + V_x = V_o$$

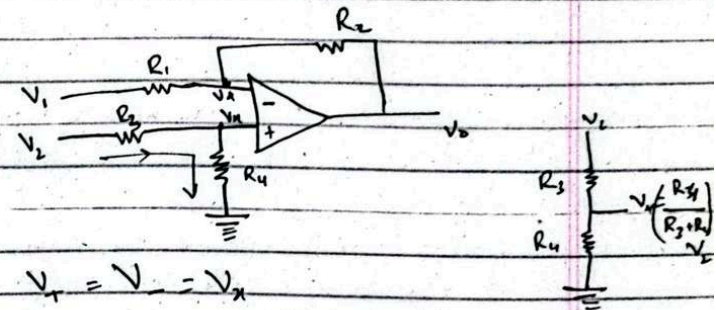
$$10V_x - 9V_3 = V_o$$

$$2(5V_x) - 9V_3 = V_o$$

$$V_o = 6V_1 + 4V_2 - 9V_3$$

lasty

Single Op-amp Difference Amp



$$V_+ = V_- = V_x$$

$$\Rightarrow \frac{V_1 - V_2}{R_3} + \frac{V_x - 0}{R_4} = 0$$

$$\frac{V_x - V_2}{R_3} + \frac{V_x}{R_4} = 0$$

$$\left(\frac{R_3 + R_4}{R_3 \cdot R_4} \right) V_x = \frac{V_2}{R_3}$$

$$V_x = \left(\frac{R_4}{R_3 + R_4} \right) V_2 \quad \text{--- (1)}$$

$$\Rightarrow \frac{V_x - V_1}{R_1} + \frac{V_x - V_o}{R_2} = 0$$

$$\frac{V_x - V_1}{R_1} + \frac{V_x}{R_2} = \frac{V_o}{R_2}$$

$$V_x \left(\frac{R_1 + R_2}{R_1 \cdot R_2} \right) - \frac{V_1}{R_1} = \frac{V_o}{R_2}$$

$$V_o = V_x \frac{R_2}{R_1} \left(\frac{R_1 + R_2}{R_1 \cdot R_2} \right) - \left(\frac{R_2}{R_1} \right) V_1$$

~~Expected~~ diff. amp. $V_o = \beta(V_2 - V_1)$
 same weighted

$$V_o = \left(1 + \frac{R_2}{R_1}\right)V_2 - \left(\frac{R_2}{R_1}\right)V_1 \quad (2)$$

Put (1) in (2)

$$V_o = \left(1 + \frac{R_2}{R_1}\right)\left(\frac{R_4}{R_3 + R_4}\right)V_2 - \left(\frac{R_2}{R_1}\right)V_1 \quad (3)$$

$$= \gamma V_2 - \beta V_1 \quad \text{unbalanced weighted}$$

$$\Rightarrow \gamma = \beta$$

$$\left(1 + \frac{R_2}{R_1}\right)\left(\frac{R_4}{R_3 + R_4}\right) = \frac{R_2}{R_1}$$

$$\left(\frac{R_1 + R_2}{R_1}\right)\left(\frac{R_4}{R_3 + R_4}\right) = \frac{R_2}{R_1}$$

$$\left(\frac{R_1}{R_2} + 1\right)\left(\frac{R_4}{R_3 + R_4}\right) = 1$$

$$\frac{R_1}{R_2} + 1 = \frac{R_3 + R_4}{R_4} \quad \text{Add 1}$$

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

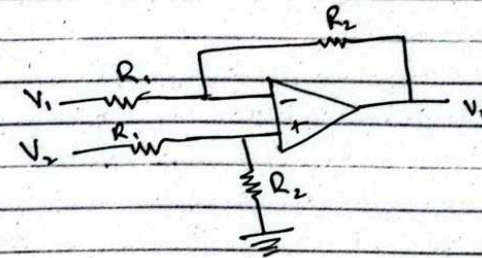
$$\frac{R_4}{R_3} = \frac{R_2}{R_1}$$

$$\Rightarrow R_1 = R_3$$

$$\Rightarrow R_2 = R_4$$

Applying $\frac{R_4}{R_3} = \frac{R_2}{R_1}$ in (3)

$$V_o = \frac{R_2}{R_1}(V_2 - V_1)$$



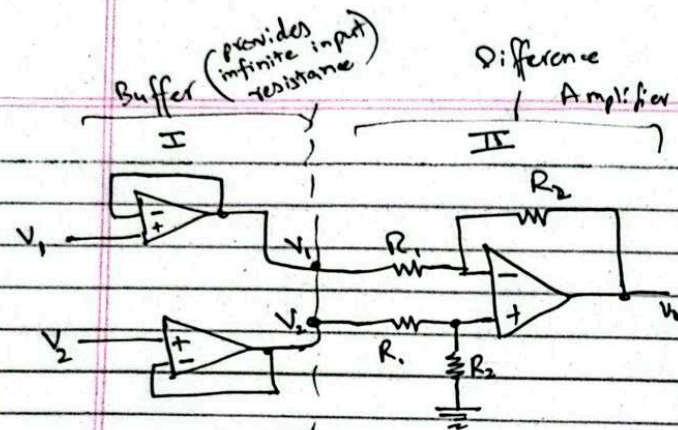
$$\frac{R_4}{R_3} = \frac{R_2}{R_1} \Rightarrow \text{Ratio must be same}$$

Proble $R_i \uparrow$ for high Gain, R_f not feasible
 $R_f \checkmark$ $G \checkmark$ $R_i \downarrow \times$
 $R_f \checkmark$ $G \times$ $R_i \checkmark$

For R_1 & R_2 , ratios must be exactly same.

Ob; R_i : large G : no compromise R_f : Feasible

To remove dependence from R_i , we add unity gain (buffer) voltage follower. It gives infinite input resistance



$$V_0 = \frac{R_2}{R_1} (V_2 - V_1)$$

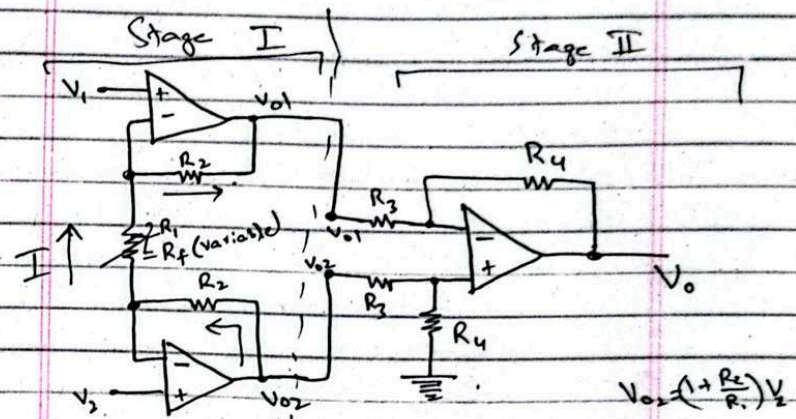
$$V_0 = G_1 (V_2 - V_1)$$

We divide total gain G_1 and also get some from Buffer. Let eg $G_1 = 100$

Buffer ($G_1 = 10$) } Gain = 10

Buffer ($G_1 = 10$) } Total Gain = 100

Instrumentation Amplifier:-



$$I = \frac{V_2 - V_1}{2R_1}$$

$$V_{02} = \left(1 + \frac{R_2}{R_1}\right) V_2$$

$$V_{01} = \left(1 + \frac{R_2}{R_1}\right) V_1$$

$$\Rightarrow V_{02} - V_{01} = I (R_2 + 2R_1 + R_2)$$

$$V_{02} - V_{01} = 2I (2R_2 + 2R_1)$$

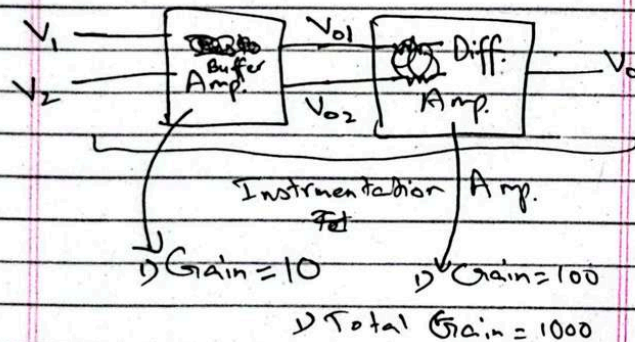
$$V_{02} - V_{01} = \left(\frac{V_2 - V_1}{2R_1}\right) (2R_2 + 2R_1)$$

$$= \left(1 + \frac{2R_2}{2R_1}\right) (V_2 - V_1)$$

$$\Rightarrow V_0 = \frac{R_4}{R_3} (V_{02} - V_{01})$$

$$= \left(\frac{R_4}{R_3}\right) \left(1 + \frac{2R_2}{R_1}\right) (V_2 - V_1)$$

This amplifier removes dependency on R_f . For equal ratio, we changed R_3 and R_1 in ratio. To overcome this, we use instrumentation amplifier. To increase gain, we change R_F ($2R_1$) only in first stage.



2) $G_1 = 5$

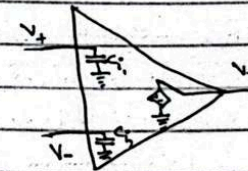
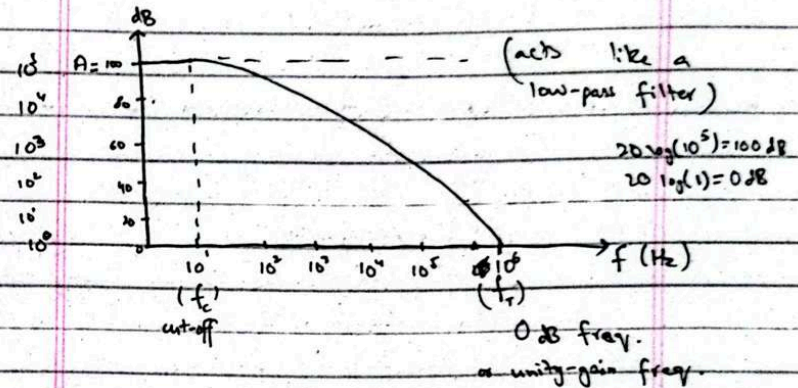
2) $G_2 = 5$

2) $TG = 25$

Transfer func: LPF $\Rightarrow \frac{\omega_0}{s + \omega_0} = \frac{\omega_0}{\omega_0(\frac{s}{\omega_0} + 1)} = \frac{1}{1 + \frac{s}{\omega_0}}$

HPF $\Rightarrow \frac{s}{s + \omega_0}$

Frequency Dependence of Open-Loop Op-Amp Gain



Ideal: $A(s) = A_0$ (constant gain throughout)

Real: $A(s) = \frac{A_0}{1 + \frac{s}{\omega_0}}$

$s = j\omega$

Low pass filter

$$A(\omega) = \frac{A_0}{1 + j\left(\frac{\omega}{\omega_0}\right)}$$

For large frequency $\Rightarrow$ at large values of ω : $1 \ll \frac{\omega}{\omega_0}$

$$A(\omega) = \frac{A_0}{j\frac{\omega}{\omega_0}} = \frac{A_0 \omega_0}{j\omega} = -j \frac{A_0 \omega_0}{\omega}$$

$$|A(\omega)| \propto \frac{1}{\omega}$$

$$|A(\omega)| = \frac{A_o \omega_o}{\omega}$$

open-loop

∴ Magnitude of Amp. gain at any frequency 'ω' can be evaluated as shown:-

Q) What is the amp. open-loop gain at unity-gain frequency?

$$|A(\omega)| = 1$$

Q) Hence, what is unity-gain frequency?

$$1 = \frac{A_o \omega_o}{\omega_f} \Rightarrow \omega_f = A_o \omega_o$$

$$2\pi f_f = A_o 2\pi f_o$$

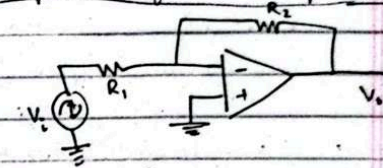
$$f_f = A_o f_o$$

$$= 10^5 \times 10^1$$

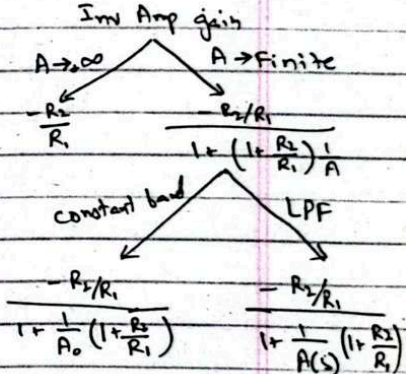
$$= 10^6$$

Frequency Dependence of Closed-loop G-amp

Inv. Amp



$$G = \frac{v_o}{v_i} = \frac{-R_2/R_1}{1 + (1 + \frac{R_2}{R_1}) \frac{1}{A}}$$



$$G = \frac{v_o}{v_i} = \frac{-R_2/R_1}{1 + \frac{1}{A(s)} \left(1 + \frac{R_2}{R_1}\right)}$$

$$\frac{v_o}{v_i} = \frac{-R_2/R_1}{1 + \left[\frac{\left(1 + \frac{R_2}{R_1}\right)}{\frac{A_o}{1 + \frac{s}{\omega_o}}}\right]}$$

$$\begin{aligned}\frac{V_o}{V_i} &= \frac{-R_2/R_1}{1 + \left(1 + \frac{R_2}{R_1}\right) \left(1 + \frac{s}{A_o \omega_b}\right)} \\ &= \frac{-R_2/R_1}{1 + \left(1 + \frac{R_2}{R_1}\right) \left(\frac{1}{A_o} + \frac{s}{A_o \omega_b}\right)} \\ &= \frac{-R_2/R_1}{1 + \frac{1}{A_o} \left(1 + \frac{R_2}{R_1}\right) + \frac{s}{A_o \omega_b} \left(1 + \frac{R_2}{R_1}\right)} \\ &= \frac{-R_2/R_1}{1 + \frac{1}{A_o} \left(1 + \frac{R_2}{R_1}\right) + \frac{s}{\left(\frac{A_o \omega_b}{1 + \frac{R_2}{R_1}}\right)}}\end{aligned}$$

$$\Rightarrow A_o = 10^5 \text{ (very large)}$$

$$\Rightarrow \frac{V_o(s)}{V_i(s)} = \frac{-R_2/R_1}{1 + \frac{s}{\frac{A_o \omega_b}{1 + \frac{R_2}{R_1}}}}$$

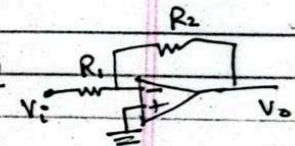
Open-Loop Op-Amp



$$A(s) = \frac{A_o}{1 + \frac{s}{\omega_b}} \quad ; \quad \omega_T = A_o \omega_b \text{ (unity gain freq.)}$$

$$|A(\omega)| = \frac{A_o \omega_b}{\omega} \quad \omega_b = \text{cut off freq.}$$

Closed-Loop Op-amp (inv.)



$$G(s) = \frac{-R_2/R_1}{1 + \frac{s}{\left(\frac{A_o \omega_b}{1 + \frac{R_2}{R_1}}\right)}} \Leftrightarrow \frac{G_o}{1 + \frac{s}{\omega_Q}} \quad \omega_Q = A_o \omega_b \left(1 + \frac{R_2}{R_1}\right)$$

$$\omega_Q^* = \frac{A_o \omega_b}{1 + \frac{R_2}{R_1}} \quad \omega_Q = A_o \omega_b \left(1 + \frac{R_2}{R_1}\right)$$

cutoff frequency of inv. amplifier

$$|G(\omega)| = \left(\frac{R_2}{R_1}\right) \left(\frac{A_o \omega_b}{1 + \frac{R_2}{R_1}}\right) \frac{1}{\omega}$$

$$\Rightarrow \frac{G_o \omega_Q}{\omega}$$

$$\omega_2 = \omega_Q G_o$$

$$\omega_2^* = \left(\frac{A_o \omega_b}{1 + \frac{R_2}{R_1}}\right) \left(\frac{R_2}{R_1}\right)$$

unity gain freq.

Closed-Loop Op-Amp (Non-Inv)

$$G(s) = \frac{1 + \frac{R_2}{R_1}}{1 + \frac{s}{\frac{A_0 \omega_0}{1 + \frac{R_2}{R_1}}}}$$

$$G(s) = \frac{1 + \frac{R_2}{R_1}}{1 + \frac{s}{\frac{A_0 \omega_0}{1 + \frac{R_2}{R_1}}}}$$

$$G(s) = \frac{1 + \frac{R_2}{R_1}}{1 + \frac{s}{\frac{A_0 \omega_0}{1 + \frac{R_2}{R_1}}}}$$

$$= \frac{1 + \frac{R_2}{R_1}}{1 + \frac{s}{\frac{A_0 \omega_0}{1 + \frac{R_2}{R_1}}}}$$

$$= \frac{1 + \frac{R_2}{R_1}}{1 + \frac{s}{\frac{A_0 \omega_0}{1 + \frac{R_2}{R_1}}}}$$

$$= \frac{1 + \frac{R_2}{R_1}}{1 + \frac{s}{\frac{A_0 \omega_0}{1 + \frac{R_2}{R_1}}}}$$

$\omega_R = \omega_0$ Cutoff freq. of inv and non-inv is same.

$$G(s) = \frac{1 + \frac{R_2}{R_1}}{1 + \frac{s}{\frac{A_0 \omega_0}{1 + \frac{R_2}{R_1}}}} = G_0$$

$$G_0 = 1 + \frac{R_2}{R_1} ; \omega_R = \frac{A_0 \omega_0}{1 + \frac{R_2}{R_1}}$$

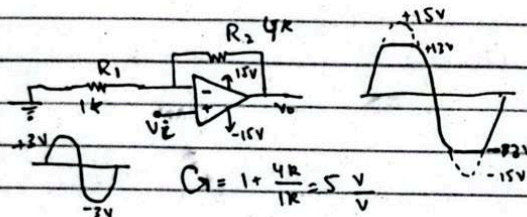
$$|G(\omega)| = \frac{G_0 \omega_R}{\omega} = \frac{(1 + \frac{R_2}{R_1}) (\frac{A_0 \omega_0}{1 + \frac{R_2}{R_1}})}{\omega}$$

$$\text{Cutoff-freq} \Rightarrow \omega_R = \frac{A_0 \omega_0}{1 + \frac{R_2}{R_1}}$$

$$\text{Unity-gain freq} \Rightarrow \omega_x = G_0 \omega_R = (1 + \frac{R_2}{R_1}) (\frac{A_0 \omega_0}{1 + \frac{R_2}{R_1}})$$

DC imperfections of op-amps

1) % voltage saturation



2) % current limit

→ Max op current limited to $\pm 20mA$

→ To amplify current, BJT is added to output pin of op-amp.

3) SLEW Rate

~~If the~~ SR (slew rate)

If the input signal applied is changing or varying faster than the value of SR, the op-amp will not comply.

$$V_i = V_m \sin(\omega t)$$

$$\frac{dV_i}{dt} = V_m \omega \cos \omega t$$

$$\left. \frac{dV_i}{dt} \right|_{\max} = V_m \omega$$

$$SR \geq \left. \frac{dV_i}{dt} \right|_{\max}$$

$$SR \geq \omega V_i$$

If ωV_i is less than SR, it is safe to use circuit. safe operation

$SR < \omega V_i$: unsafe operation
signal distorted

$$\text{As } SR = \omega_{\max} V_{i\max}$$

$$SR = 2\pi f_{\max} V_{i\max}$$

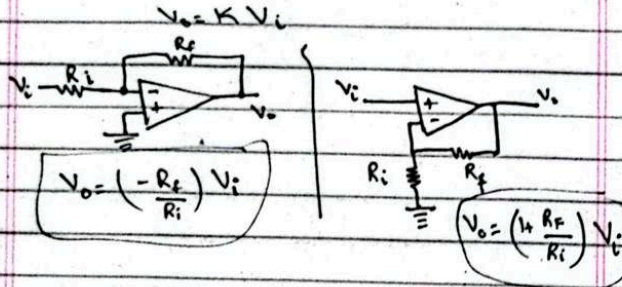
$$\Rightarrow f_{\max} = \frac{SR}{2\pi V_{i\max}}$$

$$\Rightarrow V_{i\max} = \frac{SR}{2\pi f_{\max}}$$

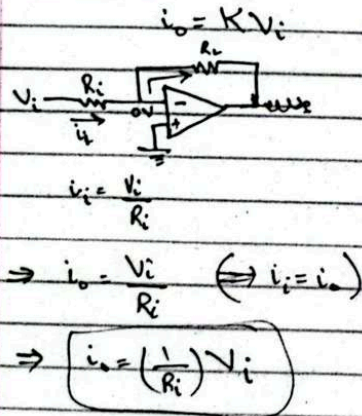
$\Rightarrow V_o$ is voltage across R_L .
 I_o is current through R_L .

Dependent Current & Voltage Sources

1) VC-VS

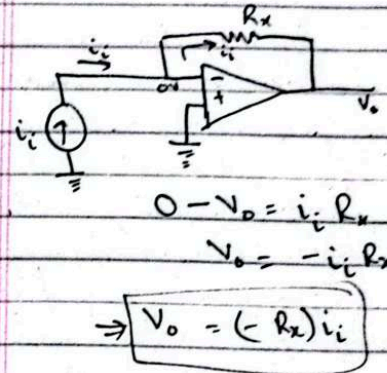


2) VC-CS



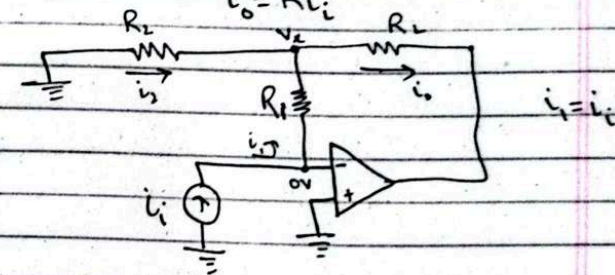
3) CC-VS

$$V_o = K i_i$$



4) CC-CS

$$i_o = K i_i$$



$$0 - V_x = i_i R_i$$

$$V_x = -i_i R_i$$

$$0 - V_x = i_i R_i$$

$$V_x = -i_i R_i$$

$R_i = \frac{V_x}{i_i}$

$R_i = \frac{-i_i R_i}{i_i}$

$R_i = -R_i$

From ① & ②

$$-i_1 R_1 = -i_2 R_2$$

$$i_2 = \left(\frac{R_1}{R_2}\right) i_1 \quad \text{--- ③}$$

At $i_1 + i_2 = i_o$

Put ③

$$i_1 + \left(\frac{R_1}{R_2}\right) i_1 = i_o$$

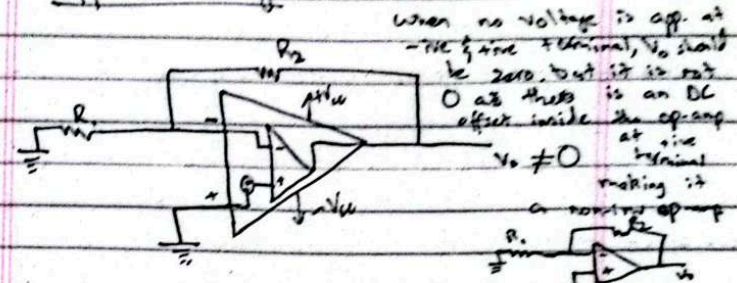
$$\Rightarrow i_o = \left(1 + \frac{R_1}{R_2}\right) i_1$$

$$\Rightarrow i_o = \left(1 + \frac{R_1}{R_2}\right) i_i$$

(continued)

DC Imperfections:-

4) Offset Voltage

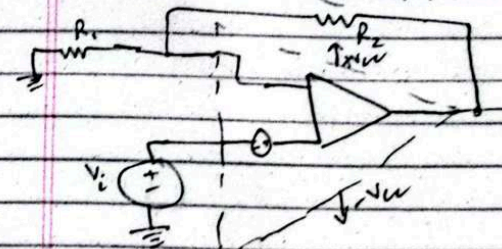


$\Rightarrow$ Non-inverting amp.

$$V_o = \left(1 + \frac{R_2}{R_1}\right) V_i$$

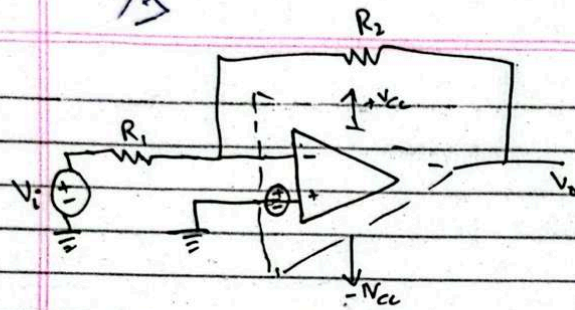
$$\Rightarrow V_o = V_{os}$$

$$V_o = \left(1 + \frac{R_2}{R_1}\right) V_{os} \neq 0$$



$$V_o = \left(1 + \frac{R_2}{R_1}\right) (V_i + V_{os})$$

$$= \left(1 + \frac{R_2}{R_1}\right) V_i + \left(1 + \frac{R_2}{R_1}\right) V_{os}$$



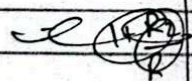
Ignoring V_{os}

$$V_o = \left(-\frac{R_2}{R_1}\right) V_i$$

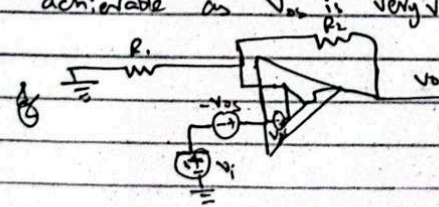
Ignoring V_i

$$V_o = \left(1 + \frac{R_2}{R_1}\right) V_{os}$$

$$V_o = \left(-\frac{R_2}{R_1}\right) V_i + \left(1 + \frac{R_2}{R_1}\right) V_{os}$$



⇒ To cancel V_{os} , we give $-V_{os}$ but it is not achievable as V_{os} is very very small.



$$V_o = \left(1 + \frac{R_2}{R_1}\right) (V_i + V_{os} + V_{os})$$

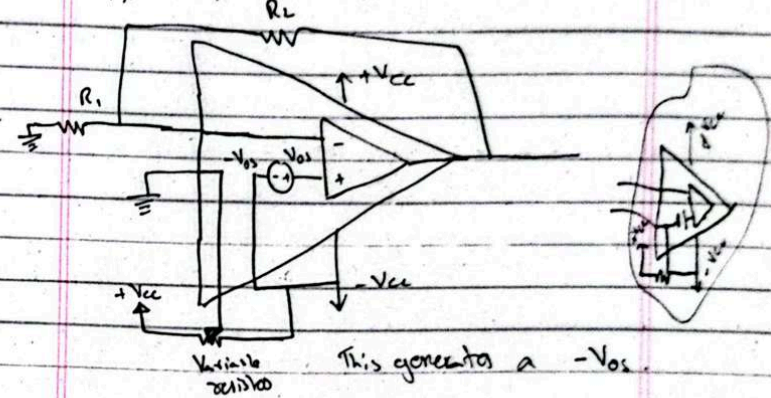
$$= \left(1 + \frac{R_2}{R_1}\right) (V_i)$$

As V_{os} is very small, this is not suitable

⇒ $(V_{os}) V_{offset}$ is always a DC signal



⇒ Offset pin is used in this case

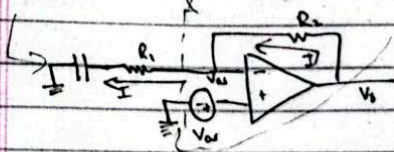


1st Remedy:-

Use a potentiometer at 'offset' pin of op-amp to introduce an equal but opposite voltage in series with V_{os} .

2nd Remedy (Partial Remedy/Cost Effective):-

↳ This only works if V_i is AC.



As I flows through circuit, we place capacitor as it does not allow DC current to pass through.

$$V_o - V_{os} = I R_1$$

$$\Rightarrow I = 0$$

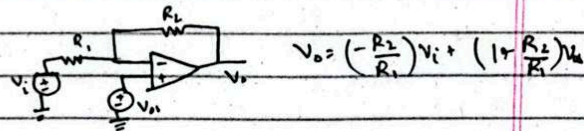
$$V_o - V_{os} = 0$$

$$V_o = V_{os}$$

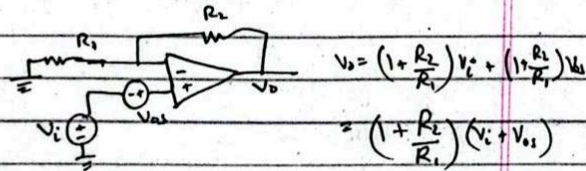
Before when capacitor is placed

$$V_o = \left(1 + \frac{R_2}{R_1}\right) V_i$$

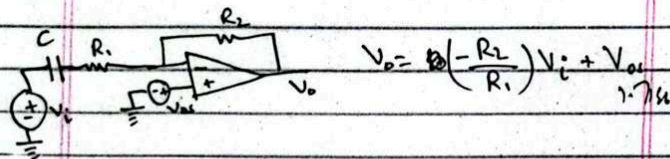
⇒ Inv. with V_{os}



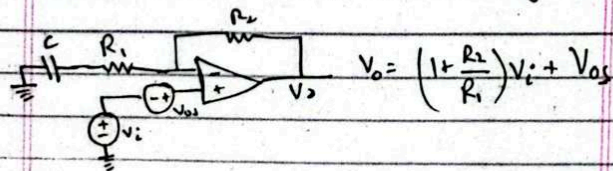
⇒ Non-Inv. with V_{os}



⇒ Inv. with V_{os} & Partial Remedy

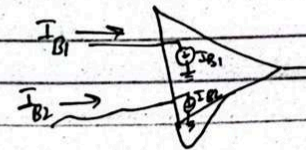


⇒ Non-Inv. with V_{os} & Partial Remedy



5) Input Bias Current

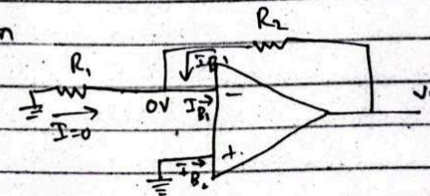
In real opamp, some non-zero current I_{B1} & I_{B2} flow through terminals. (inside)
Although I_{B1} & I_{B2} are of order nano-amperes, but they are a problem



Avg. Input bias current $\Rightarrow I_B = \frac{I_{B1} + I_{B2}}{2} \approx 100 \text{ nA}$

Input offset current $\Rightarrow I_{os} = |I_{B1} - I_{B2}| \approx 1 \text{ nA (or)}$

Problem

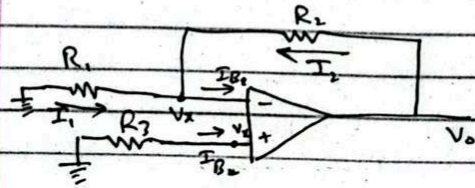


$$\Rightarrow V_o - 0 = I_{B1} R_2$$

$$V_o = I_{B1} R_2$$

Remedy

Put R_3 at non-inv. terminal



For simplicity: $I_{B1} = I_{B2} = I_B$

$$\Rightarrow 0 - V_x = I_B R_3$$

$$V_x = -I_B R_3$$

$$\Rightarrow 0 - V_x = I_1 R_1$$

$$I_1 = I_B \left(\frac{R_3}{R_1} \right)$$

By KCL,

$$I_1 + I_2 = I_B$$

$$I_2 = I_B - I_1$$

$$I_2 = I_B - I_B \left(\frac{R_3}{R_1} \right)$$

$$\Rightarrow V_0 - V_x = I_2 R_2$$

$$V_0 = V_x + I_2 R_2$$

$$V_0 = -I_B R_3 + \left(I_B - I_B \left(\frac{R_3}{R_1} \right) \right) R_2$$

$$= I_B \left(-R_3 + R_2 - \frac{R_2 R_3}{R_1} \right)$$

$$\Rightarrow V_0 = I_B \left(R_2 - R_3 \left(1 + \frac{R_2}{R_1} \right) \right)$$

Let Put $V_0 = 0$

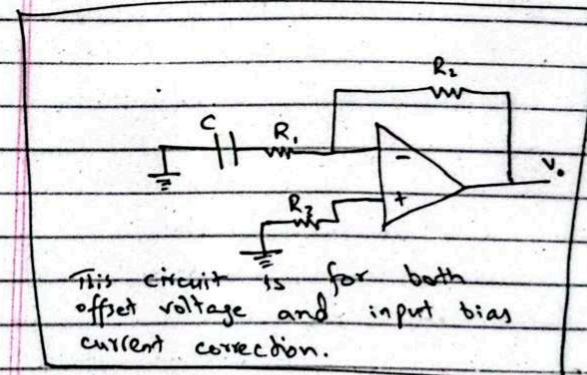
$$0 = I_B \left(R_2 - R_3 \left(1 + \frac{R_2}{R_1} \right) \right)$$

$$0 = R_2 - R_3 \left(1 + \frac{R_2}{R_1} \right)$$

$$R_2 = R_3 \left(1 + \frac{R_2}{R_1} \right)$$

$$R_3 = \frac{R_1 R_2}{R_1 + R_2}$$

$$\Rightarrow R_3 = R_1 \parallel R_2$$

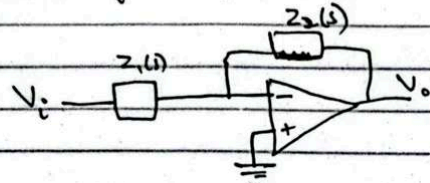


This circuit is for both offset voltage and input bias current correction.

Capacitor Laplace $\left(\frac{1}{Cs}\right)$

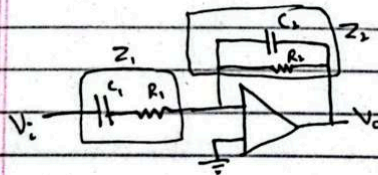
Inductor Laplace Ls

Inverting Conf. with General Impedance



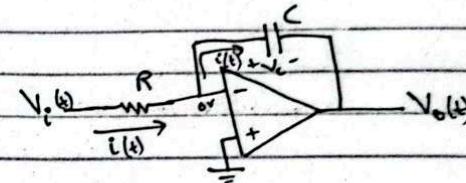
$$G(s) = \frac{V_o(s)}{V_i(s)} = - \frac{Z_2(s)}{Z_1(s)}$$

Gain or transfer func.



$$\begin{aligned} \frac{V_o(s)}{V_i(s)} &= - \frac{Z_2(s)}{Z_1(s)} = - \frac{R_2 \parallel \frac{1}{Cs}}{R_1 + \frac{1}{Cs}} = - \frac{R_2 \left(\frac{1}{Cs} \right)}{R_1 + \frac{1}{Cs}} \\ &= - \frac{R_2 / Cs}{\left(R_1 + \frac{1}{Cs} \right) \left(R_1 + \frac{1}{Cs} \right)} \end{aligned}$$

Inverting Integrator Op-Amp:-



$$\Rightarrow i(t) = \frac{V_i(t)}{R}$$

$$\Rightarrow i(t) = C \frac{dV_c(t)}{dt}$$

$$\frac{dV_c(t)}{dt} = \frac{i(t)}{C}$$

$$\int \frac{dV_c(t)}{dt} dt = \int \frac{i(t)}{C} dt$$

$$V_c(t) = \frac{1}{C} \int i(t) dt + V_c(0)$$

$$\Rightarrow V_c(0) = 0$$

$$V_c(t) = \frac{1}{C} \int i(t) dt$$

$$= \frac{1}{C} \int \frac{V_i(t)}{R} dt$$

$$= \frac{1}{RC} \int V_i(t) dt$$

$$V_o(t) = - \frac{1}{RC} \int V_i(t) dt$$

0-Volt $V_c(t)$
 $V_o(t) = -V_c(t)$

$\frac{1}{s} \rightarrow$ 

$$\frac{V_o(s)}{V_i(s)} = - \frac{Z_2(s)}{Z_1(s)} = - \frac{\frac{1}{Cs}}{R}$$

$$= - \frac{1}{RCs}$$

$$V_o(s) = -\frac{1}{RC_3} \left(\frac{V_i(s)}{s} \right)$$

Laplace inverse on bs

$$V_o(t) = -\frac{1}{RC} \int V_i(t) dt$$

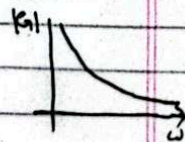
$$G(s) = - \frac{1}{RCs}$$

$$G(j\omega) = -\frac{1}{jR\omega} = \frac{j}{R\omega}$$

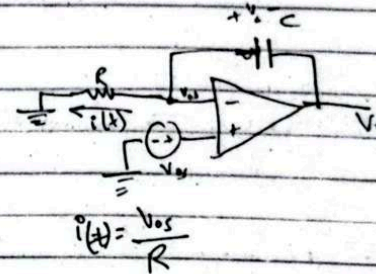
$$= 0 + j\left(\frac{1}{R\omega}\right)$$

$$|G| = \sqrt{0^2 + \left(\frac{1}{RC\omega}\right)^2} = \frac{1}{RC\omega}$$

$$\Delta G = \tan^{-1} \left(\frac{R\omega}{0} \right) = 90^\circ$$



Impact of V_{os} on Inverter Behavior



$$\Rightarrow i(t) = C \frac{dV_c}{dt}$$

$$V_c = \frac{1}{C} \int i(t) dt$$

$$= \frac{1}{C} \int \frac{V_{os}}{R} dt$$

$$V_c = \frac{1}{RC} \int_{t=0}^{\infty} V_{in} dt$$

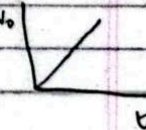
$$V_o - V_{o1} = V_c$$

~~$$V_o = V_c + V_d$$~~

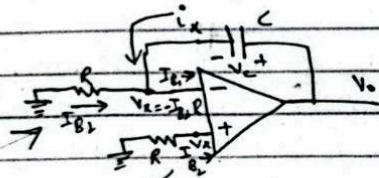
$$V_o = V_o + \frac{1}{RC} \int_0^t V_o dt$$

$$= V_a + \frac{1}{R_E} (V_{OS} * t)$$

$$V_o = V_{os} + \left(\frac{V_{os}}{RC} \right) t \quad (y = mt + c)$$



Impact of I_B on Integrator's Behavior



$$\Rightarrow 0 - V_x = I_{B2} R$$

$$V_x = -I_{B2} R$$

$$i = \frac{0 - V_x}{R}$$

$$i = \frac{0 - (-I_{B2} R)}{R}$$

$$= \frac{I_{B2} R}{R}$$

$$i = I_{B2}$$

$$i_x + I_{B2} = I_{B1}$$

$$i_x = I_{B1} - I_{B2}$$

$$V_c = \frac{1}{C} \int i_x(t) dt$$

$$V_c = \frac{1}{C} \int (I_{B1} - I_{B2}) dt$$

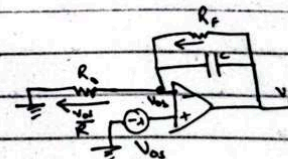
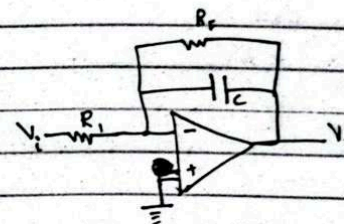
$$V_c = \left(\frac{I_{B1} - I_{B2}}{C} \right) t$$

$$V_c = V_o - V_x$$

$$V_o = V_c + V_x$$

$$V_o = -I_{B2} R + \left(\frac{I_{B1} - I_{B2}}{C} \right) t$$

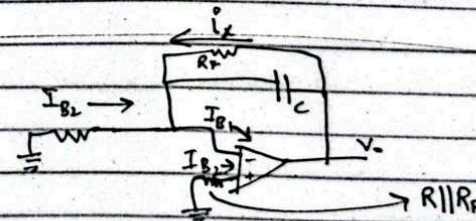
Remedy



$$V_o = V_{os} + \left(\frac{V_{os}}{R} \right) R_f$$

$$= \left(1 + \frac{R_f}{R} \right) V_{os}$$

$$V_o = V_{os}$$

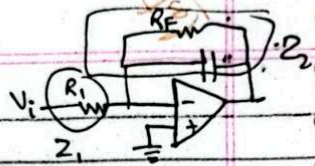


$$V_o - (-I_{B2} R) = R i_x$$

$$V_o = I_{B2} R + R i_x$$

$$V_o = I_{B2} R + R (I_{B1} - I_{B2})$$

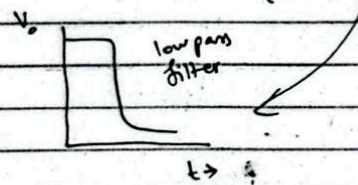
page 4 of 5
Date: / /



$$G(s) = \frac{V_o(s)}{V_i(s)} = - \frac{Z_2(s)}{Z_1(s)}$$

$$= - \frac{\frac{1}{C_F} \parallel R_F}{R_i}$$

$$G(s) = - \frac{R_F/R_i}{1 + s(C_F R_F)}$$

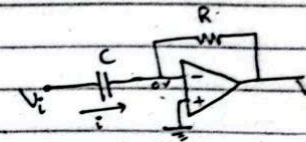


R_F : very small

ideal

$$G(s) = - \frac{1}{RCs}$$

OP-AMP Differentiator



$$i = C \frac{dV_i}{dt}$$

$$= C \frac{dV_i}{dt}$$

$$\Rightarrow 0 - V_o = iR$$

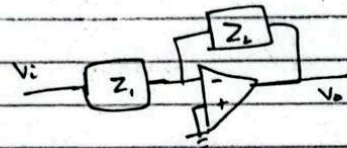
$$V_o = -iR$$

$$V_o = -C \frac{dV_i}{dt} R$$

$$V_o = -RC \frac{dV_i}{dt}$$

d	-1	100 pF	10	10 pA	1000 x 10 ³	10
C	-1	10 pF	10 pA	1000 x 10 ³	10	10
m	-1	10 pF	10 pA	1000 x 10 ³	10	10
n	-1	10 pF	10 pA	1000 x 10 ³	10	10
		10 x 10 ³	100000	1000 x 10 ³	10 pA	10
		0.010				10

General Impedance Differentiator



$$\frac{V_o(s)}{V_i(s)} = -\frac{Z_2}{Z_1}$$

$$\frac{V_o(s)}{V_i(s)} = -\frac{R}{\frac{1}{Cs}}$$

$$\frac{V_o}{V_i} = -RCs$$

$$V_o(s) = -RC (V_i(s) s)$$

Laplace Inverse

(noise amplifier)
(high pass filter)

$$V_o(t) = -RC \left(\frac{dV_i}{dt} \right)$$

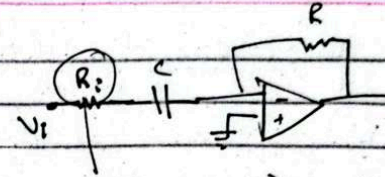
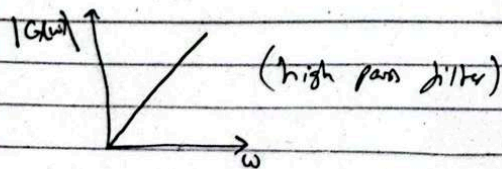
$$\frac{V_o(s)}{V_i(s)} = -RCs$$

$$s = j\omega$$

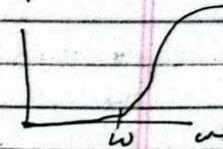
$$G(\omega) = \frac{V_o(\omega)}{V_i(\omega)} = -RC(j\omega) = 0 - j(RC\omega)$$

$$|G(\omega)| = (RC)\omega$$

$$\angle G(\omega) = \tan^{-1} \left(\frac{-RC\omega}{0} \right) = -90^\circ$$



This resistor is placed
to cancel noise



$$\frac{V_o}{V_i} = -\frac{R}{R_i + \frac{1}{Cs}} \left(-\frac{Z_2}{Z_1} \right)$$

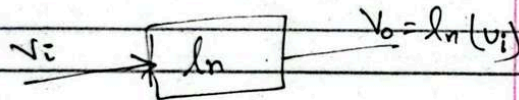
$$= -\frac{R/R_i}{1 + \frac{1}{RCs}} \rightarrow$$

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$$\frac{V_o}{V_i} = -RCs$$

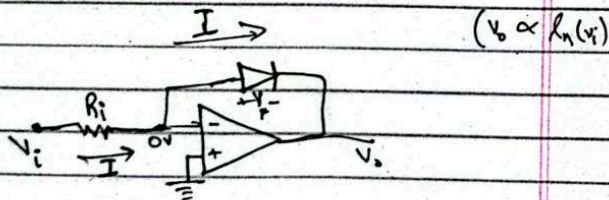
Non-Linear Operations of Op-Amp

Logarithmic AMP



⇒ All p-n junction / semi-conductor devices exhibit an exponential behavior because there is a depletion region.

⇒ Log amplifier is an inverting amp. The feedback resistor is replaced with any p-n junction



$$I = \frac{V_i}{R_i}$$

Diode I-V Eq:-

$$I = I_s \left(e^{\frac{V_F}{\eta V_T}} - 1 \right)$$

I = forward current

→ depends of surrounding conditions (P, T, q)

I_s = reverse saturation current

V_F = Voltage across diode

V_T = Thermal voltage = ηV_T

$\eta = 1$ for Si
 $\eta = 2$ for Ge

Assume: $V_F > \eta V_T$

$$\Rightarrow e^{\frac{V_F}{\eta V_T}} \gg 1$$

Hence neglect '1'

$$I = I_s e^{\frac{V_F}{\eta V_T}}$$

Take $\ln$ on bs

$$\ln I = \ln(I_s e^{\frac{V_F}{\eta V_T}})$$

$$\ln I = \ln I_s + \ln \left(e^{\frac{V_F}{\eta V_T}} \right)$$

$$\ln I = \ln I_s + \frac{V_F}{\eta V_T}$$

$$V_F = (\ln I - \ln I_s) \eta V_T$$

$$= \eta V_T \left(\ln \left(\frac{V_i}{R_i} \right) - \ln I_s \right)$$

R_T = Thermistor resistance
 R_i = very small (0.1 or 1 Ω)

$$\frac{I_2}{V_i/R}$$

$$\Rightarrow 0 - V_o = V_T$$

$$V_o = -V_T$$

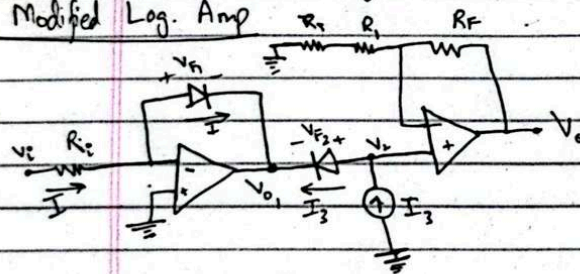
$$V_o = -\eta V_T \left(\ln \left(\frac{V_i}{R_i} \right) - \ln I_s \right)$$

Problems:

1) $\ln(I_s)$ needs to be removed

2) $V_o \propto \eta V_T$

Modified Log. Amp



$$\Rightarrow V_{o1} = -\eta V_T \left(\ln \left(\frac{V_i}{R_i} \right) - \ln(I_s) \right)$$

$$\Rightarrow V_{F2} = +\eta V_T \left(\ln I_3 - \ln I_s \right)$$

$$\Rightarrow V_2 - V_{o1} = V_{F2}$$

$$V_2 = V_{F2} + V_{o1}$$

$$= -\eta V_T \ln \frac{V_i}{R_i} + \eta V_T \ln I_s$$

$$+ \eta V_T \ln I_3 - \eta V_T \ln I_s$$

$$= -\eta V_T \ln \frac{V_i}{R_i} + \eta V_T \ln I_3$$

$$= \eta V_T \left(\ln I_3 - \ln \frac{V_i}{R_i} \right)$$

$$= \eta V_T \left(\ln \left(\frac{V_i}{R_i} \right) - \ln I_s \right)$$

$$= -\eta V_T \left(\ln \frac{V_i}{R_i} - \ln I_s \right)$$

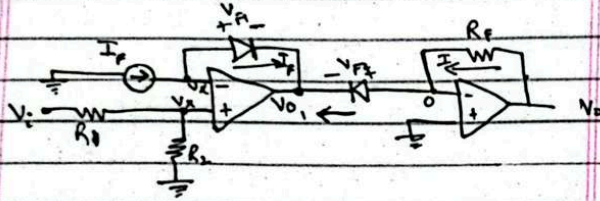
$$= -\eta V_T \ln \left(\frac{V_i}{I_s R_i} \right)$$

$$V_o = \left(1 + \frac{R_F}{R_i + R_T} \right) V_2$$

$$V_o = - \left(1 + \frac{R_F}{R_i + R_T} \right) \eta V_T \ln \left(\frac{V_i}{I_s R_i} \right)$$

I_3 is constant
 I_s depends on surrounding

Anti-Log Amp



$$V_x = \left(\frac{R_2}{R_1 + R_2} \right) V_i$$

$$V_{F1} = +\eta V_T (\ln I_F - \ln I_S)$$

$$\Rightarrow V_x - V_{01} = V_{F1}$$

$$\textcircled{1} V_{01} = V_x - V_{F1}$$

$$V_{01} = \left(\frac{R_2}{R_1 + R_2} \right) V_i - \eta V_T (\ln I_F - \ln I_S) \quad \textcircled{1}$$

$$V_{F2} = +\eta V_T (\ln I - \ln I_S)$$

$$\Rightarrow I = \frac{V_0}{R_F}$$

$$\Rightarrow 0 - V_{01} = V_{F2}$$

$$V_{01} = -V_{F2} \quad \textcircled{2}$$

Substitute ① in ②

$$-V_{F2} = V_x - V_{F1}$$

$$-\eta V_T (\ln I - \ln I_S) = \left(\frac{R_2}{R_1 + R_2} \right) V_i - \eta V_T (\ln I_F - \ln I_S)$$

$$\left(\frac{R_2}{R_1 + R_2} \right) V_i = \eta V_T (\ln I_F - \ln I)$$

$$\ln \left(\frac{I_F}{I} \right) = \frac{1}{\eta V_T} \left(\frac{R_2}{R_1 + R_2} \right) V_i$$

$$\ln \left(\frac{I}{I_F} \right) = -\frac{1}{\eta V_T} \left(\frac{R_2}{R_1 + R_2} \right) V_i$$

$$\frac{I}{I_F} = \ln^{-1} \left(-\frac{1}{\eta V_T} \left(\frac{R_2}{R_1 + R_2} \right) V_i \right)$$

$$I = I_F \ln^{-1} \left(-\frac{1}{\eta V_T} \left(\frac{R_2}{R_1 + R_2} \right) V_i \right)$$

$$\frac{V_0}{R_F} = I_F \ln^{-1} \left(-\frac{1}{\eta V_T} \left(\frac{R_2}{R_1 + R_2} \right) V_i \right)$$

$$V_0 = I_F R_F \ln^{-1} \left(-\frac{1}{\eta V_T} \left(\frac{R_2}{R_1 + R_2} \right) V_i \right)$$

$$V_0 = K_1 \ln^{-1} \left(-\frac{V_i}{R_1} \right)$$

$$K_1 = I_F R_F$$

Not in book

$$\frac{1}{\eta V_T} \left(\frac{R_2}{R_1 + R_2} \right)$$

$$R_1 \gg R_2$$

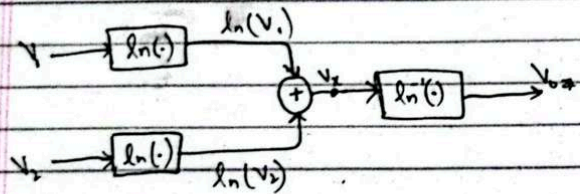
$$\frac{1}{\eta V_T} \left(\frac{R_2}{R_1} \right)$$

$$\frac{1}{\eta V_T} \left(\frac{R_2}{R_1} \right)$$

$$\eta = 1 \text{ for silicon}$$

$$V_i \rightarrow \ln(\cdot) \rightarrow V_o = -K \ln(V_i)$$

$$V_i \rightarrow \ln^{-1}(\cdot) \rightarrow V_o = K \ln^{-1}(V_i)$$

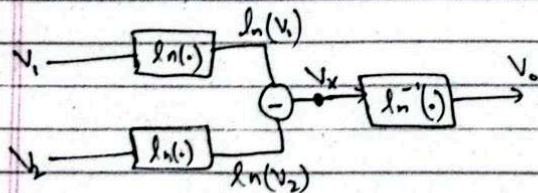


$$V_x = \ln(V_1) + \ln(V_2)$$

$$V_x = \ln(V_1 V_2)$$

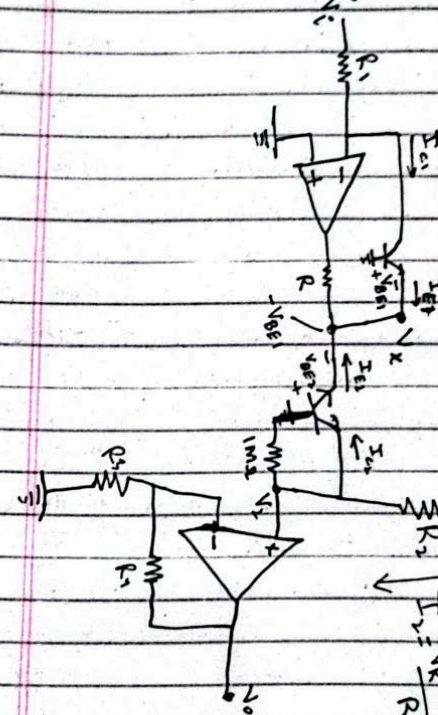
$$V_o = \ln^{-1}(V_x) = \ln^{-1}(\ln(V_1 V_2))$$

$$V_o = V_1 V_2 \quad \text{analog multiplier}$$

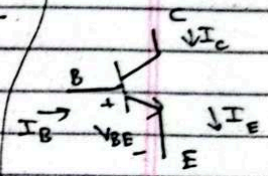


$$V_o = \frac{V_1}{V_2}$$

Modified Log Amp



BJT z-tics



$$I_B = I_C$$

$$I_C + I_B = I_E$$

$$V_{BE} \geq 0.7$$

$$V_B - V_E \geq 0.7$$

$$V_B \geq V_E + 0.7$$

$$I_E = I_S e^{\frac{V_{BE}}{V_T}}$$

$$\Rightarrow V_{BE} = V_T \ln\left(\frac{I_E}{I_S}\right)$$

$$\Rightarrow 0 - V_x = V_{BE1}$$

$$V_x = -V_{BE1}$$

$$\Rightarrow I_{E1} \frac{V_i}{R_1} \approx I_{E1}$$

$$V_{BE1} = V_T \ln\left(\frac{I_{E1}}{I_S}\right)$$

$$V_{BE2} = V_T \ln\left(\frac{I_{E2}}{I_S}\right)$$



$$\Rightarrow V_2 - (-V_{BE1}) = V_{BE1}$$

$$V_2 + V_{BE1} = V_{BE1}$$

$$V_2 = V_{BE2} - V_{BE1}$$

$$\begin{aligned} V_2 &= V_T \ln\left(\frac{I_{E2}}{I_1}\right) - V_T \ln\left(\frac{I_{E1}}{I_1}\right) \\ &= V_T \ln\left(\frac{I_{E2}}{I_{E1}}\right) \end{aligned}$$

$\Rightarrow V_2$ is so much smaller than V_R that it is neglected. $V_2 \ll V_R$

$$I_2 = \frac{V_R - V_2}{R_1} \approx \frac{V_R}{R_1}$$

$$I_{E2} = I_2 = \frac{V_R}{R_1}$$

$$\begin{aligned} I_{E2} &\approx I_{C2} \\ I_{E2} &= \frac{V_R}{R_1} \end{aligned}$$

$$\Rightarrow V_2 = V_T \ln\left(\frac{V_R/R_1}{V_i/R_1}\right)$$

$$V_2 = V_T \ln\left(\frac{V_R R_1}{V_i R_1}\right)$$

$$V_2 = -V_T \ln\left(\frac{V_i R_1}{V_R R_1}\right)$$

$$\text{Put } \frac{R_2}{R_1 V_R} = 1$$

Suppose $V_R = 10V$, $R_1 = 10K$, $R_2 = 100K$

$$V_2 = -V_T \ln(V_i)$$

$$\Rightarrow V_o = \left(1 + \frac{R_4}{R_3}\right) V_2$$

$$= -\left(1 + \frac{R_4}{R_3}\right) V_T \ln(V_i)$$

$$\frac{R_4}{R_3} \gg 1$$

$$V_o = -\left(\frac{R_4}{R_3}\right) V_T \ln(V_i)$$

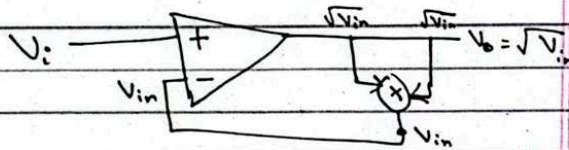
Put $R_3 = R_T$ to cancel out V_T (thermistor)

$$V_o = -\frac{R_4}{R_T} \left(\frac{V_T}{R_T}\right) \ln(V_i)$$

$$V_o = -K \ln(V_i)$$

Square Root Extraction:-

$$V_i \rightarrow \sqrt{(\cdot)} \rightarrow V_o = \sqrt{V_i}$$

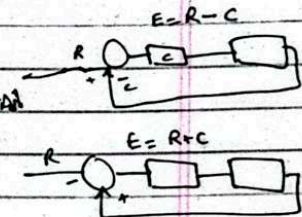


Sine Wave Generator (Oscillator)

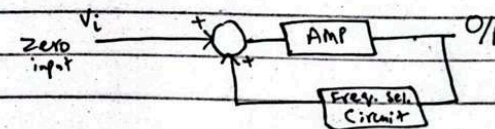
Oscillator: Oscillators generate a finite output signal by using an amplifier followed by frequency selective circuit connected in positive feedback loop with no NO input signal

⇒ -ve feedback is stable as error converges to 0.

→ +ve feedback diverges upto ∞ .



Sinusoidal Oscillation



Suppose $V_i = x_i$

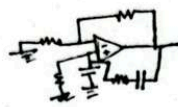
x_i = reservoir of infinite sinoid frequencies

$$x_i = M_1 \sin \omega(t) + M_2 \sin(2\omega t) + M_3 \sin(3\omega t) + \dots + M_n \sin(n\omega t)$$

magnitude of $M_1, M_2, M_3, \dots, M_n$ is very very small.
of the order 10^{-30} .

Microcontroller time control:
crystal oscillator with two
paper capacitor. This
generates a clock pulse.

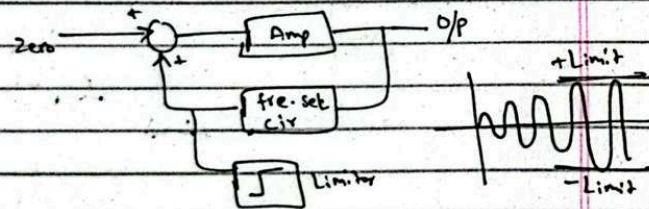
Thermal noise of the resistor



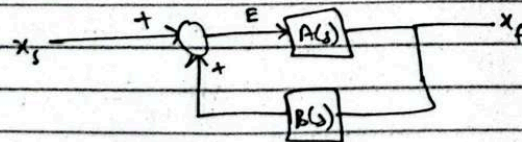
$$\text{transfer func} = \frac{\text{output}}{\text{input}}$$

$$\begin{aligned} x_s + 0 &\rightarrow Ax_s \rightarrow ABx_f \\ x_s + Ax_f &\rightarrow Ax_s + A^2\beta x_f \rightarrow A\beta \end{aligned}$$

Improvement



Thermal noise of the resistor:-



$$E = x_s + B(s)x_f \quad \text{--- (1)}$$

$$x_f = A(s)E \quad \text{--- (2)}$$

Put (1) in (2)

$$x_f = A(s) [x_s + B(s)x_f]$$

$$x_f = A(s)x_s + A(s)B(s)x_f$$

$$x_f [1 - A(s)B(s)] = A(s)x_s$$

If we apply a zero input, the output would be a finite signal of frequency f_0 if and only if $A(\omega_0)B(\omega_0) = 1$ holds at frequency f_0 .

$$\frac{x_f}{x_s} = \frac{A(s)}{1 - A(s)B(s)} \quad \text{--- (i)}$$

$$\Rightarrow \frac{x_f}{x_s} = \frac{x_f}{0} = \infty \quad \text{--- (ii)}$$

From (i) & (ii)

$$1 - A(s)B(s) = 0$$

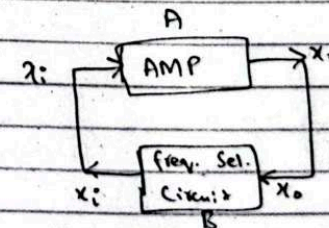
$$A(s)B(s) = 1 \quad \text{BARK-HOUSEN'S CRITERIA}$$

$$\begin{aligned} A(\omega_0)B(\omega_0) &= 1 \\ \omega_0 &= 2\pi f_0 \end{aligned}$$

so if we apply a zero input, the output would be a finite signal of frequency f_0 if and only if.

$$A(\omega_0)B(\omega_0) = 1 \quad ; \quad \omega_0 = 2\pi f_0$$

holds at the frequency f_0 .



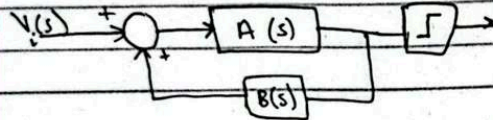
$$x_i = Bx_o$$

$$x_o = Ax_i$$

$$x_o = ABx_o$$

$$AB = 1$$

Wein-Bridge Oscillator:-

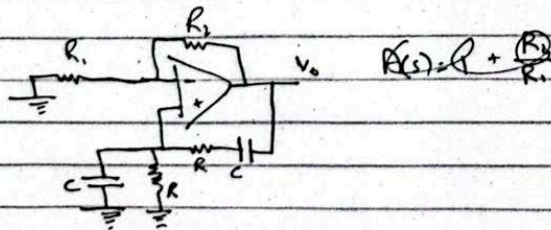
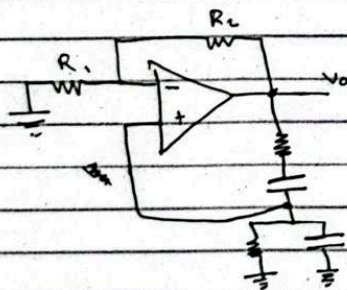


Bark-Hausen Criteria

to get a finite ~~sign~~ ^{sine} signal at f_0

$$A(\omega_0)B(\omega_0) = 1$$

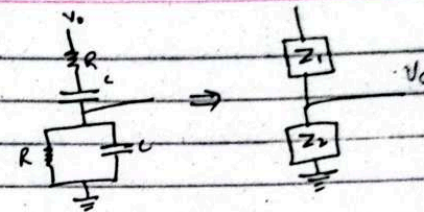
$$\omega_0 = 2\pi f_0$$



$$B(s) = 1 + \frac{R_2}{R_1}$$

$$\frac{1}{R_{eq}} = \frac{1}{R} + \frac{1}{\frac{1}{sC}} \quad \frac{R}{1+RCs}$$

$$\frac{1}{R} + \frac{sC}{1+RCs}$$



$$B(s) = \frac{Z_2}{Z_1 + Z_2}$$

$$= \frac{(R \parallel \frac{1}{sC})}{(R \parallel \frac{1}{sC}) + (R \parallel \frac{1}{sC})}$$

$$B(s) = \frac{1}{3 + (RC)s + \left(\frac{1}{RC}\right)s}$$

$$A(s) = 1 + \frac{R_2}{R_1}$$

$$A(s)B(s) = 1$$

$$A(s)B(s) = \frac{1 + \frac{R_2}{R_1}}{3 + (RC)s + \frac{1}{(RC)s}}$$

Put $s = j\omega$

$$A(\omega)B(\omega) = \frac{1 + \frac{R_2}{R_1}}{3 + j(RC\omega) + \frac{1}{j(RC\omega)}}$$

$$= \frac{1 + \frac{R_2}{R_1}}{3 + j(RC\omega) - \frac{1}{RC\omega}}$$

B-H Criteria: To get a real finite sine wave of freq f_0 , $A(\omega_0)B(\omega_0) = 1$ at $f = f_0$

$$A(\omega_0)B(\omega_0) = \frac{1 + \frac{R_2}{R_1}}{3 + j\left(RC\omega_0 - \frac{1}{RC\omega_0}\right)}$$

If we make the imaginary part = zero,
then this whole thing becomes Real.

To reach the system real:

$$RC\omega_0 - \frac{1}{RC\omega_0} = 0$$

$$RC\omega_0 = \frac{1}{RC\omega_0}$$

$$\omega_0^2 = \frac{1}{(RC)^2} \Rightarrow \omega_0 = \frac{1}{RC}$$

$$\omega_0 = \frac{1}{RC} \quad (\text{filter design})$$

$$A(\omega_0)B(\omega_0) = 1$$

$$\frac{1 + \frac{R_2}{R_1}}{3 + j\left(RC\omega_0 - \frac{1}{RC\omega_0}\right)} = 1$$

$$\frac{1 + \frac{R_2}{R_1}}{3} = 1$$

$$1 + \frac{R_2}{R_1} = 3$$

$$\Rightarrow \frac{R_2}{R_1} = 2$$

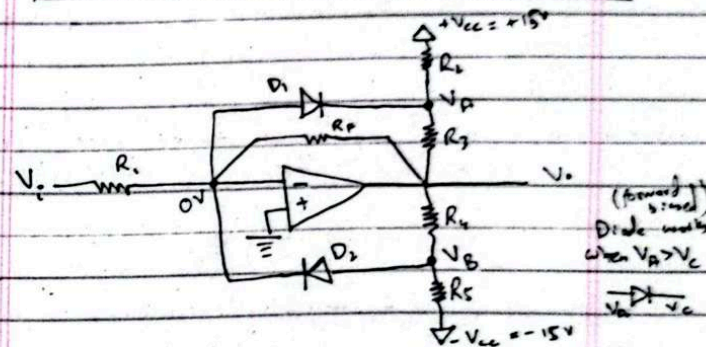
$$\Rightarrow \frac{R_3}{R_1} \geq 2$$

Practically

$$\text{Filter: } \omega_0 = \frac{1}{RC}$$

$$\text{Amp: } \frac{R_3}{R_1} \geq 2$$

Limiting Circuit of Amplitude Control:

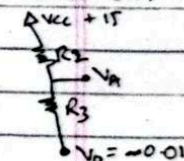


→ V_i is very small (close to 0) but +ve

→ $V_o = -\left(\frac{R_f}{R_i}\right)V_i$

→ V_o is also very small but negative let ≈ -0.01

For small V_i , diodes are open circuit



→ V_i gradually increases

→ V_o will be more & more -ive

→ V_o continues to decrease in magnitude until $V_A = -0.7V \rightarrow$ at this point, diode D_1 closes (forward bias)

→ As soon as D_1 closes (short circuit), R_3 becomes parallel to R_f (Hence, gain decreases)

$$V_o = -\left(\frac{R_3 || R_f}{R_i}\right)V_i$$

$$\frac{R_f \gg R_3 || R_f}{R_i} \gg G_1$$

For negative cycle of sine wave,
(small) $-V_i \Rightarrow +V_o$ (small)

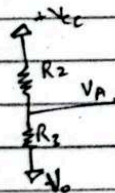
when $-V_i$ increases $\Rightarrow +V_o$ increases
Hence V_o increases

V_o increases until $V_o = +0.7V$
(sat -ive)

at this point D_2 closes (short circuit) (fb)

$$V_o = - \left(\frac{R_F || R_4}{R_1} \right) V_i$$

Hence, gain decreases
due to D_2 .

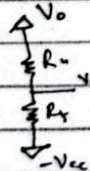


Principle of superposition

$$V_A = \left(\frac{R_3}{R_2 + R_3} \right) (V_{cc}) + \left(\frac{R_2}{R_2 + R_3} \right) (V_o)$$

$$V_A = -V_o, V_o = L_-$$

$$\Rightarrow L_- = -V_{cc} \left(\frac{R_3}{R_2} \right) - V_o \left(1 + \frac{R_3}{R_2} \right)$$

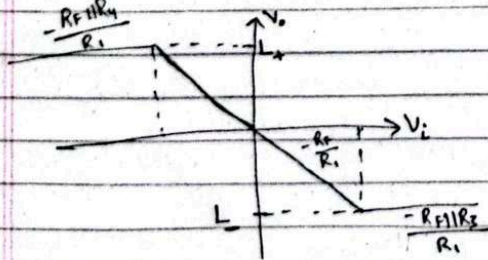


Prin. of superposition

$$V_B = \left(\frac{R_4}{R_4 + R_5} \right) (V_{cc}) + \left(\frac{R_5}{R_4 + R_5} \right) (V_o)$$

$$V_B = +V_o, V_o = L_+$$

$$\Rightarrow L_+ = +V_{cc} \left(\frac{R_4}{R_5} \right) + V_o \left(1 + \frac{R_4}{R_5} \right)$$



MVB

Multi-Vibrator CCTs for Signal Generation

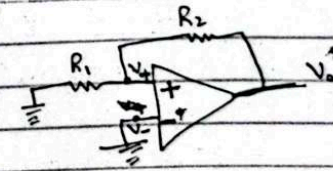
- 1) Bistable MVB (2 stable state)
- 2) Monostable MVB (1 stable state)
- 3) Astable MVB (No stable state)

1) Bistable MVB:

- Bistability: DC amplifier in a positive feedback loop.

- Open loop gain: $A\beta > 1$

- Feedback contains a voltage divider of R_1 & R_2



1) $V_- \rightarrow 0V$

2) No external input voltage

3) Electrical noise in circuit introduces

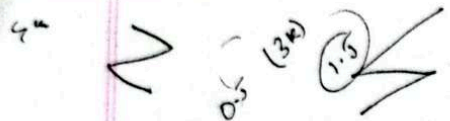
A small +ive voltage at V_+

Hence, $V_o' = A(V_+ - V_-)$

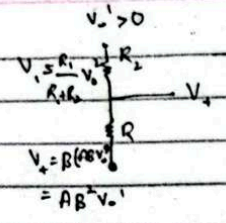
$$\Rightarrow V_o' > 0$$

Output is fed back at $V_+ = \left(\frac{R_1}{R_1 + R_2} \right) V_o$

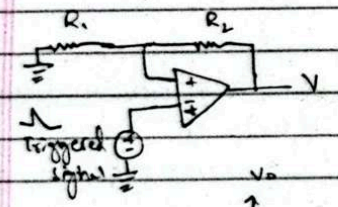
$$V_o = \beta V_o'$$



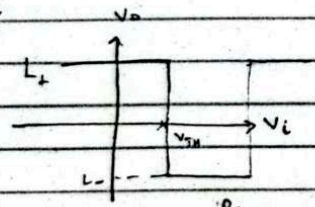
$$\begin{aligned}
 V_o^{(1)} &= ABV_o' \\
 V_o^{(2)} &= A^2B^2V_o' \\
 V_o^{(3)} &= A^3B^3V_o' \\
 V_o^{(n)} &= L_n \\
 V_o &= L_n
 \end{aligned}$$



Transfer Characteristics of Bistable MVB

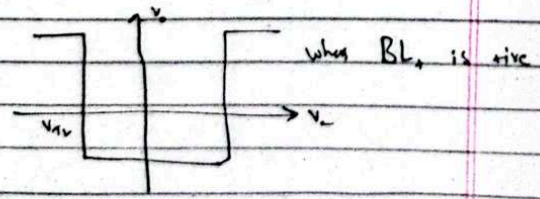


This circuit is Hysteresis comparator



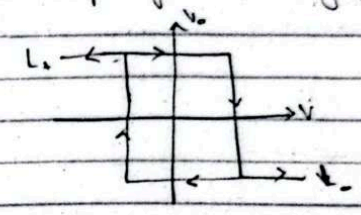
Suppose $V_o = L_+$; $V_+ = \left(\frac{R_1}{R_1 + R_2}\right) L_+ = BL_+$

- 1) $V_+ = BL_+$
- 2) $V_- = V_i$
- 3) Amp, $V_o = A(V_+ - V_-)$
- 4) Initially, $V_- = 0V$
- 5) V_i starts gradually increases



$$\begin{aligned}
 \frac{V_+}{R_1} + \frac{V_- - V_o}{R_2} &= 0 \\
 V_+ \left(\frac{1}{R_1} + \frac{1}{R_2} \right) - \frac{V_o}{R_2} &= 0 \\
 V_+ &= \frac{V_o}{R_1 + R_2}
 \end{aligned}$$

By comparing both graphs

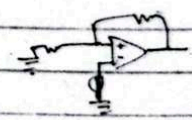
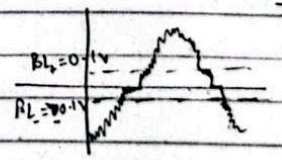


Hysteresis Loop

$$\begin{aligned}
 V_{TH} &= BL_+ \\
 V_{TL} &= BL_- \\
 B &= \frac{R_1}{R_1 + R_2}
 \end{aligned}$$

Inv. Bistable MVB / Hysteresis Comparator:-

- 1) If trigger (V_i) $> BL_+$ $\Rightarrow V_o$ goes from L_+ to L_-
- 2) If trigger (V_i) $> BL_-$ $\Rightarrow V_o$ goes from L_- to L_+

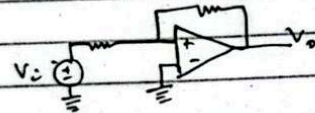
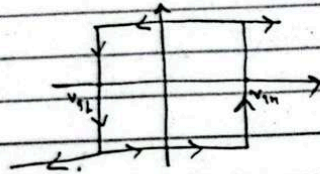


The hysteresis comparator helps to eliminate the ripple near zero crossings of signal waveform

(+ve) - (-ve) = +ve (BL_+)

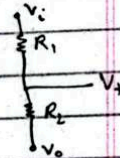
(-ve) - (+ve) = -ve (BL_-)

Non-Inv. Bistable MVB:-



By superposition

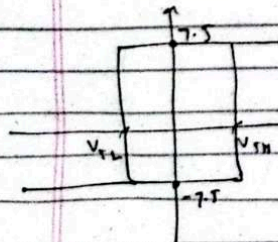
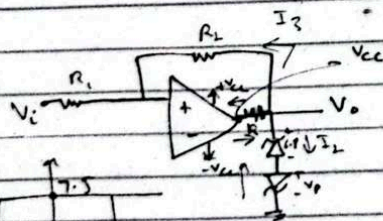
$$V_o = \frac{R_2}{R_1 + R_2} V_i + \frac{R_1}{R_1 + R_2} V_o$$



$$V_o = \left(\frac{R_2}{R_1 + R_2} \right) V_i + \left(\frac{R_1}{R_1 + R_2} \right) V_o$$

$$V_{th} = - \left(\frac{R_1}{R_2} \right) I_-$$

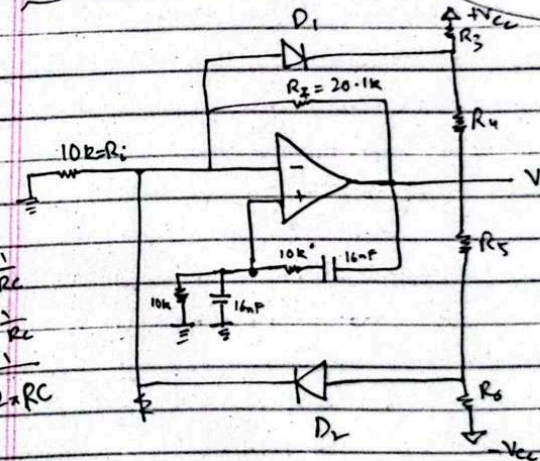
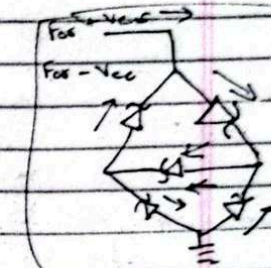
$$V_{th} = - \left(\frac{R_1}{R_2} \right) I_+$$



$$0 = V_o = V_i + V_o \Rightarrow V_o = V_i + V_o$$

$$-V_o = 0.7 + 6.8 \Rightarrow V_o = 7.5V$$

$$V_o = -7.5$$



$$\omega = \frac{1}{RC}$$

$$2\pi f = \frac{1}{RC}$$

$$f = \frac{1}{2\pi RC}$$

$$f = 994.7Hz$$

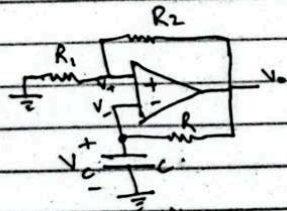
$$\approx 1kHz$$

BL₊

$$y(t) = y(\infty) - (y(\infty) - y(0))e^{-\frac{t}{RC}}$$

ASTABLE MVB (using inv. bistable config.)

- 1) No stable state
- 2) Keeps toggling b/w L₊ & L₋
- 3) Used to generate sq. waves and rectangular waves.
- 4) Uses Bistable MVD circuit with an RC circuit in -ve feedback.



$$V_o = A(V_+ - V_-)$$

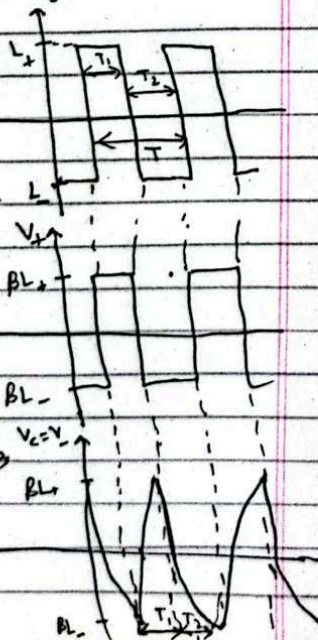
$$V_- = V_c$$

$$\text{Initially, } V_o = L_+, V_+ = BL_+$$

1) V_c will start rising exponentially

2) V₋ will start rising exponentially

3) Because V_o = L₊ is fixed, V₊ is also fixed



$$\frac{L_+(1-B)}{V_+(1-B(\frac{L_+}{L_-}))} = e^{-\frac{T_1}{RC}}$$

$$\ln\left(\frac{1-B}{1-B(\frac{L_+}{L_-})}\right) = -\frac{T_1}{RC}$$

Time Period:-

$$T_1: V_c(t) = V_c(\infty) - (V_c(\infty) - V_c(0))e^{-\frac{t}{RC}}$$

$$V_c(0) = BL_-$$

$$V_c(\infty) = L_+$$

$$\text{At } t = T_1, V_c(t) = BL_+$$

$$\Rightarrow BL_+ = L_+ - (L_+ - BL_-)e^{-\frac{T_1}{RC}}$$

$$\frac{BL_+ - L_+}{L_+ - BL_-} = -e^{-\frac{T_1}{RC}}$$

$$T_1 = RC \ln\left(\frac{1 - B(\frac{L_-}{L_+})}{1 - B}\right)$$

For symmetrical power supplies L₊ = -L₋

$$T_1 = RC \ln\left(\frac{1+B}{1-B}\right)$$

$$T_2: V_c(0) = BL_+$$

$$V_c(\infty) = L_-$$

$$\text{At } t = T_2, V_c(t) = BL_-$$

$$\Rightarrow BL_- = L_- - (L_- - BL_+)e^{-\frac{T_2}{RC}}$$

$$\frac{-BL_- + L_-}{L_- - BL_+} = e^{-\frac{T_2}{RC}}$$

$$T_2 = RC \ln\left(\frac{1 - B(\frac{L_+}{L_-})}{1 - B}\right)$$

$$B = \frac{R_1}{R_1 + R_2}$$

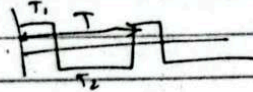
$$\Rightarrow L_+ = -L_-$$

$$T_2 = RC \ln \left(\frac{1+B}{1-B} \right)$$

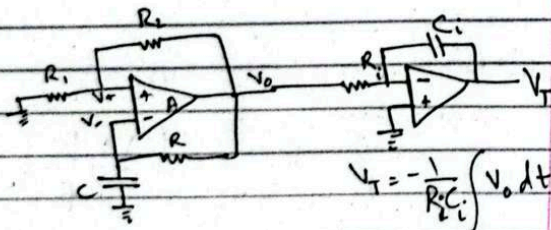
$$T_1 = T_2 \quad L_+ = -L_-$$

$$T = T_1 + T_2$$

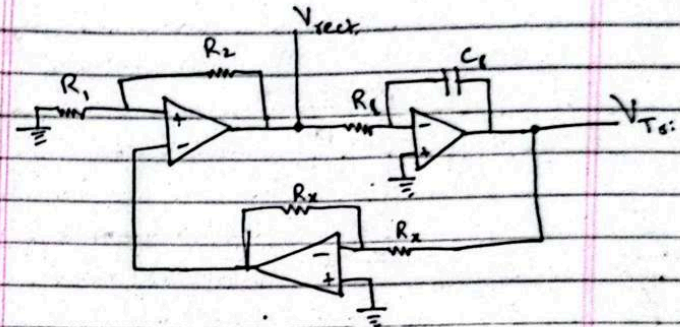
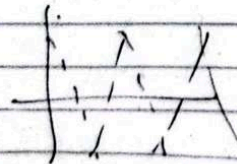
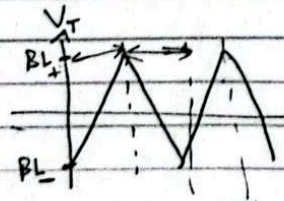
Example For rectangular wave, $L_+ \neq -L_-$



For triangular wave, join an Integrator circuit at V_o .



$$V_T = -\frac{1}{RC} \int V_o dt$$



Function part (Sq. + Δ wave) $\Rightarrow$

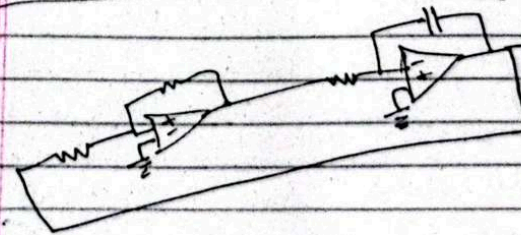
For symmetric power supply, $L_+ = -L_-$

$$T_1 = RC \left(\frac{V_{TH} - V_{TL}}{L_+} \right)$$

$$T_2 = RC \left(\frac{V_{TH} - V_{TL}}{-L_-} \right)$$

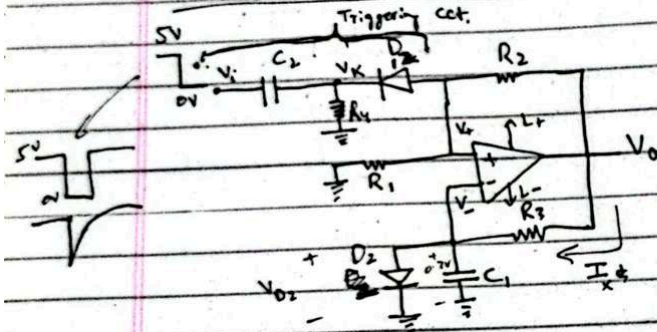
$$V_{TH} = \beta L_+$$

$$V_{TL} = \beta L_-$$

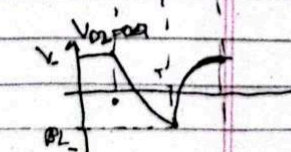
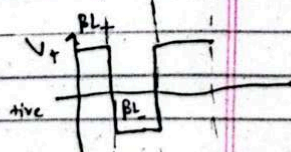
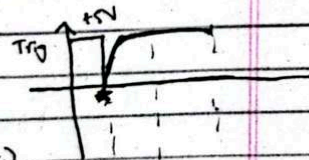


$\Rightarrow$ Low pass filter acts as integrator
 $\Rightarrow$ High pass filter acts as differentiator

Monostable MVB (Pulse Generator)



- 1) $V_0 = L_+$, $V_+ = \beta L_+$
- 2) Because $V_+ > \beta L_+$, D_2 is reverse biased. D_2 is open & D_1 is open.
- 3) For I_x flow, D_2 is fwd biased.
- 4) Hence D_2 is closed.
- 5) $V_0 = A(V_+ - V_-) = A(\beta L_+ - \beta L_-)$
- 6) When we apply a trigger; $V_+ = 0$
- 7) D_2 short ckt.
- 8) D_1 is short ckt; 0V at V_+ will travel ahead and forces itself at V_+ .
- 9) $V_0 = A(V_+ - V_-) = A(0 - V_-)$
- 10) In this duration, V_+ becomes rive again (D_1 is open).
- 11) $V_0 = A(\beta L_- - 0 \cdot 7)$
- 12) $V_0 = A(\beta L_- - V_-)$
- 13) $V_- = \beta L_-$



$$\Rightarrow y(t) = y(\infty) - (y(\infty) - y(0)) e^{-\frac{t}{RC}}$$

$$\Rightarrow V_c(t) = V_c(\infty) - (V_c(\infty) - V_c(0)) e^{-\frac{t}{R_3 C_1}}$$

$$V_c(0) = V_{D2}$$

$$V_c(\infty) = L_-$$

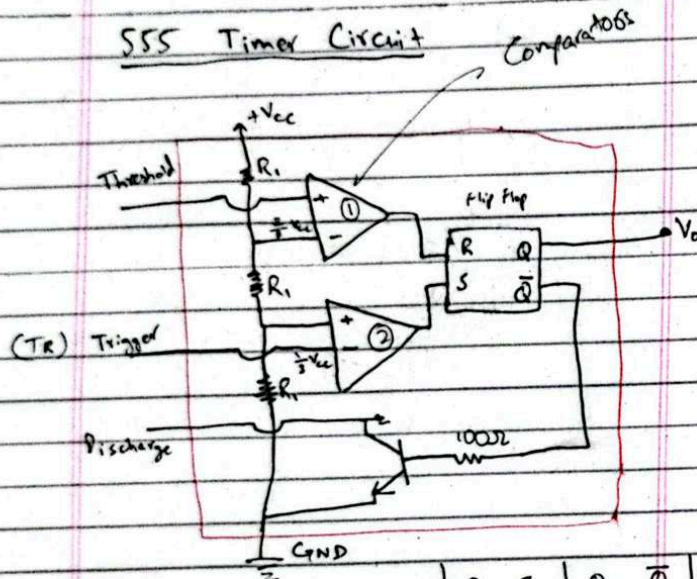
$$\text{At } t = T_c, V_c(t) = \beta L_-$$

$$\Rightarrow \beta L_- = L_- - (L_- - V_{D2}) e^{-\frac{T}{R_3 C_1}}$$

$$\Rightarrow T = R_3 C_1 \ln \left(\frac{V_{D2} - L_-}{\beta L_- - L_-} \right)$$

$$\left(\frac{R_1}{R_1 + R_2} \right)$$

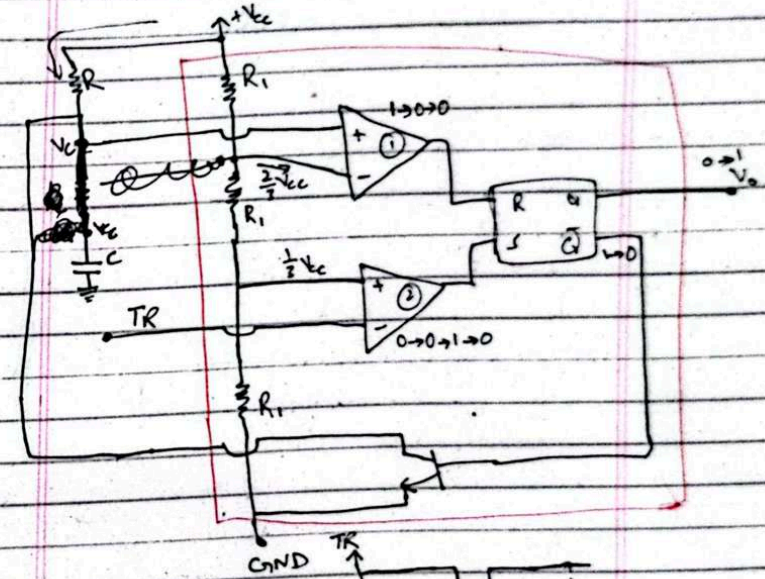
555 Timer Circuit



R	S	Q	$\bar{Q}$
0	0	previous state	
0	1	1	0
1	0	0	1
1	1	undefined	

$V_+ > V_- \Rightarrow$ Comparator output: 1
 $V_+ < V_- \Rightarrow$ Comparator output: 0

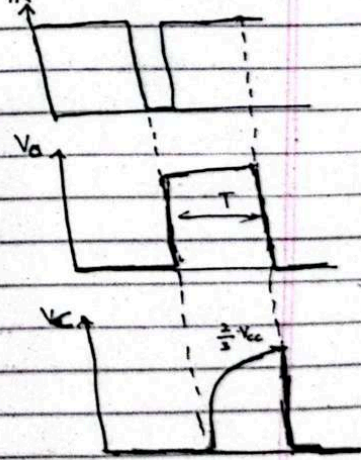
555 Timer Circuit as a Monostable MVB:-



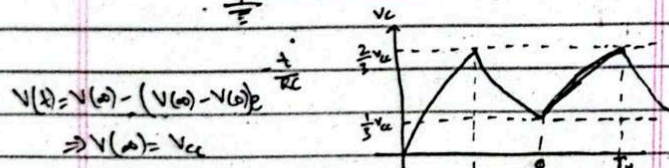
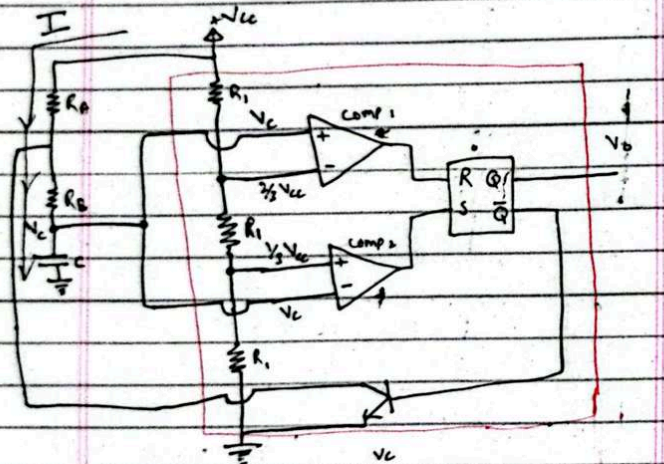
Suppose we are in
 Rest mode.

$$T = (\ln 3) RC$$

$$\Rightarrow T = 1.1 RC$$



Astable MVB using 555 timer:



$$V(t) = V(\infty) - (V(\infty) - V(0))e^{-\frac{t}{RC}}$$

$$\Rightarrow V(\infty) = V_{cc}$$

$$\Rightarrow V(0) = \frac{1}{3} V_{cc}, R = R_A + R_B$$

$$\Rightarrow \text{at } t = T_H, V(t) = \frac{2}{3} V_{cc}$$

$$\frac{2}{3} V_{cc} = V_{cc} - (V_{cc} - \frac{1}{3} V_{cc})e^{-\frac{T_H}{RC}}$$

$$\frac{2}{3} V_{cc} e^{-\frac{T_H}{RC}} = \frac{1}{3} V_{cc}$$

$$e^{-\frac{T_H}{RC}} = \frac{1}{2}$$

$$T_H = -\ln\left(\frac{1}{2}\right)RC$$

$$T_H = -\ln\left(\frac{1}{2}\right)(R_A + R_B)C$$

$$T_H = \ln 2 (R_A + R_B)C$$

$$I_{\text{charge}} = (R_A + R_B)C$$

$$I_{\text{discharge}} = R_B C$$

$$V(t) = V(\infty) - (V(\infty) - V(0))e^{-\frac{t}{RC}}$$

$$\Rightarrow V(\infty) = 0$$

$$\Rightarrow V(0) = \frac{2}{3} V_{cc}, R = R_B$$

$$\text{at } t = T_L, V(t) = \frac{1}{3} V_{cc}$$

$$\frac{1}{3} V_{cc} = 0 - (0 - \frac{2}{3} V_{cc})e^{-\frac{T_L}{R_B C}}$$

$$\frac{1}{3} V_{cc} = \frac{2}{3} V_{cc} e^{-\frac{T_L}{R_B C}}$$

$$\frac{1}{2} = e^{-\frac{T_L}{R_B C}}$$

$$\ln\left(\frac{1}{2}\right) = -\frac{T_L}{R_B C}$$

$$T_L = -\ln\left(\frac{1}{2}\right)R_B C$$

$$T_L = \ln 2 (R_B)C$$

Duty Cycle:

The percentage of total time period for which the signal is logic '1' (high).

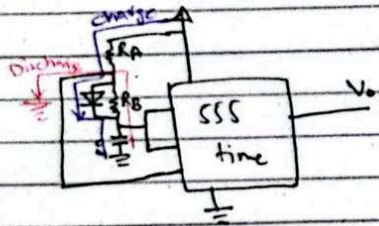
$$\Rightarrow D = \frac{T_H}{T_H + T_L} \rightarrow \text{High time}$$

$$= \frac{T_H}{T_H + T_L} \times 100\%$$

$$= \frac{R_A + R_B}{R_A + 2R_B}$$

→ In the current setting of this circuit, T_H can never be equal to T_L and hence, we can never achieve 50% duty cycle and hence no square wave using this circuit.

For 50% duty cycle,



$$D = \frac{T_H}{T} = \frac{R_A}{R_A + R_B}$$

Put $R_A = R_B$

$$D = \frac{R_A}{2R_A} = \frac{1}{2} = 50\%$$

555 Timer Circuits

1) For duty cycle $> 50\%$

$$T_1 = (\ln 2)(R_A + R_B)C$$

T_H

$$T_2 = (\ln 2)(R_B)C$$

T_L

$$T = (\ln 2)(R_A + 2R_B)C$$

$$f = \frac{1}{T}$$

$$D = \frac{R_A + R_B}{R_A + 2R_B}$$

2) For duty cycle $\leq 50\%$

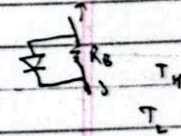
$$T_1 = (\ln 2)(R_A)C$$

$$T_2 = (\ln 2)(R_B)C$$

$$T = (\ln 2)(R_A + R_B)C$$

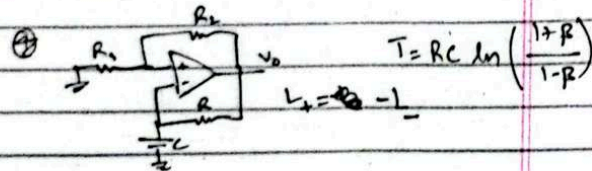
$$f = \frac{1}{T}$$

$$D = \frac{R_A}{R_A + R_B}$$

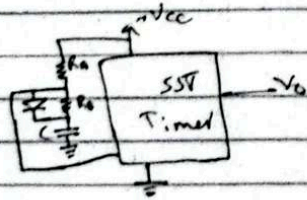


phase
 ① $C = 1 \mu F$, $R = ?$, $T = 10 \mu s$
 $\Rightarrow$ Monostable using SST
 $T = 1.1 RC$
 $R = \frac{T}{1.1 C} = \frac{(10 \mu s)}{(1.1)(1 \mu F)}$
 $R = 9.09 k\Omega$

① Propose 2-circuits for square wave
 ① Astable MVB using inv. bistable config



② Astable MVB using SST timer



① $C = 680 pF$, SST time, Rect. signal
 $\Rightarrow$ 75% duty cycle $f = 50 kHz$
 $R_A \neq R_B = ?$
 $f = \frac{1}{T} \Rightarrow T = \frac{1}{f}$
 $= 0.02 ms$

$T_H = \ln 2 (R_A + R_B) C$
 $0.75(0.02 ms) = R_A + R_B$
 $(\ln 2)(680 pF)$

$31.824 k\Omega = R_A + R_B$

$T_L = \ln 2 (R_B) C$
 $0.25(0.02 ms) = R_B$
 $(\ln 2)(680 pF)$
 $10.608 k\Omega = R_B$

$R_A = 21.216 k\Omega$

$D = 75\% = 0.75$

$\frac{R_A + R_B}{R_A + 2R_B} = \frac{3}{4}$

$\Rightarrow R_A = 2R_B$

$T = (\ln 2) (R_A + 2R_B) C$

$R_A + 2R_B = 42.43 k\Omega$

$R_A + 4R_B = 42.43 k\Omega$

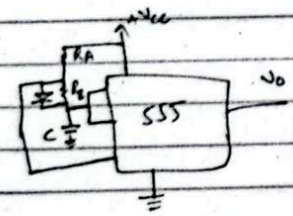
0.009 0.01 0.09 0.1

Q) SSS timer, sq. wave, $f = 10\text{KHz}$,
 $C = 1\mu\text{F}$

$T = \frac{1}{f} = 0.1\text{ms}$

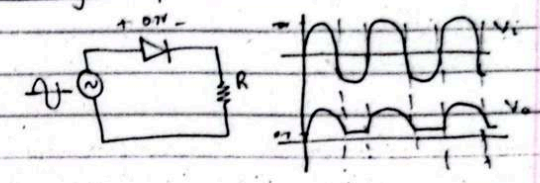
$(0.5) T = \ln 2 R_A C$
 $\Rightarrow R_A = \frac{0.5 \times 10^{-4}}{\ln 2 \times 10^{-6}} = 72.134\Omega$

$(0.5) T = \ln 2 R_B C$
 $\Rightarrow R_B = 72.134\Omega$

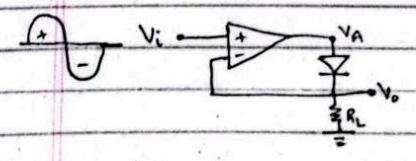


25 4 725623

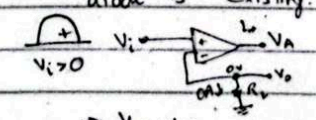
Ordinary Half-Wave Rectifier



Super-Diode



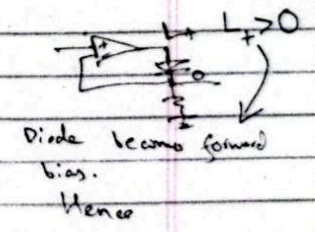
$\Rightarrow$ During the +ve half cycle ($V_i > 0$)
 $\rightarrow$ For a moment, ignores that the diode is existing.



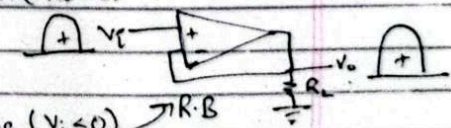
$\Rightarrow V_+ = \text{+ve}$
 $\Rightarrow V_- = 0$

$V_A = A(V_+ - V_-) = A(+\text{ve} - 0)$

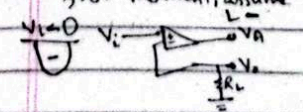
$V_A = L_+$



Diode becomes forward bias.
Hence



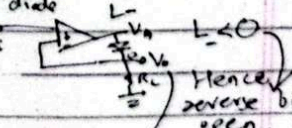
$\Rightarrow$ During -ve half cycle ($V_i < 0$)
 $\rightarrow$ For moment, assume no diode



$\Rightarrow V_+ = -\text{ive}$ $V_- = 0$

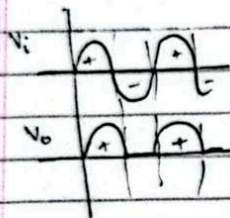
$V_A = A(V_+ - V_-) = A(-\text{ive} - 0)$

$V_A = L_-$



Hence reverse biased open.

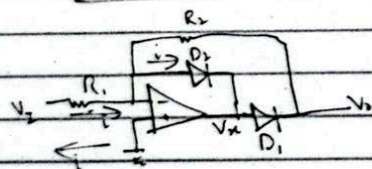
$V_o (0)$



Disadvantage

- 1) If V_i is large AC, op-amp may get -
- 2) When V_i is -ive, op-amp is saturated

Improved Circuit



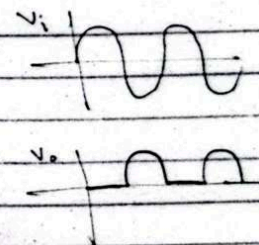
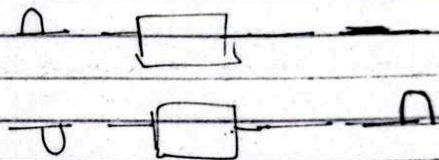
- 1) When $V_i > 0$, $i = \frac{V_i}{R_1}$ flow into -ive term
- 2) $V_- = 0V$ (ground)
- 3) Current direction ($V_i > 0$), forward bias D_1
- 4) $V_{D_1} = -0.7V$, keeps D_2 off
- 5) No current via R_2 (due to D_2 & P.) $\Rightarrow V_o = 0V$

- 1) When $V_i < 0$, $i = \frac{V_i}{R_1}$ flow out of -ive term
- 2) Due to direction of current, D_2 open
- 3) Hypothetically, V_- is much less than V_+ , so

$$V_o = A(V_+ - V_-) > 0$$

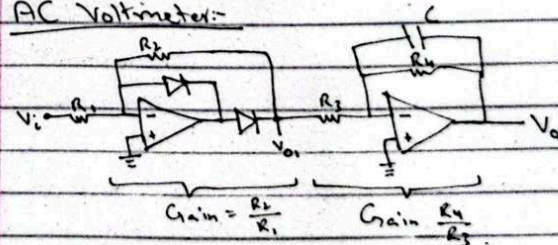
$$\Rightarrow D_1 \text{ is ON}$$

2) All the current 'i' flows through R_2
 hence $R_2 V_o = -R_1 V_i$



1) Stay in linear region

AC Voltmeter:-



2) Contains the improved precision rectifier followed by a lowpass filter.

1) The output of rectifier is half-sine wave having peak amplitude $V_p \left(\frac{R_2}{R_1} \right)$

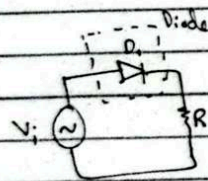
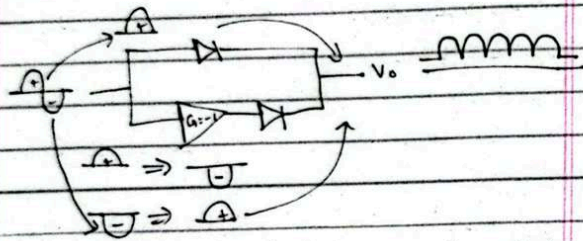
2) Also contains odd harmonics of freq. ω

1) Avg. value $\Rightarrow \frac{V_p}{2} \left(\frac{R_2}{R_1} \right)$

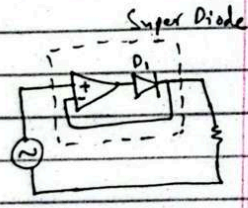
2) To reduce harmonics $\Rightarrow \frac{1}{R_3 C} \ll \omega_{min}$

2) Output voltage $\Rightarrow V_o = - \frac{V_p}{2} \left(\frac{R_2}{R_1} \right) \left(\frac{R_4}{R_3} \right)$

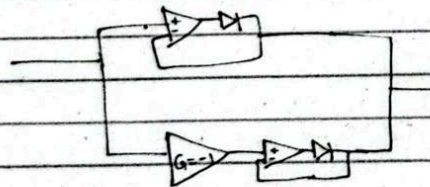
Precision - Full Wave Rectifier



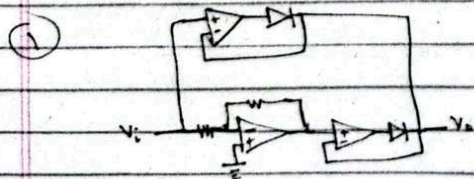
This circuit gives a half wave rectifier with a 0.7V cut off voltage



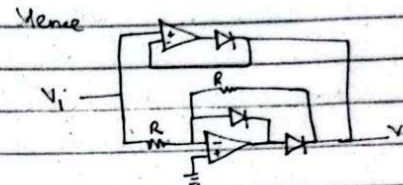
This circuit removes 0.7V cut off voltage due to op-amp



General form of Circuit

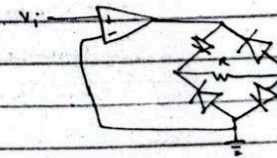
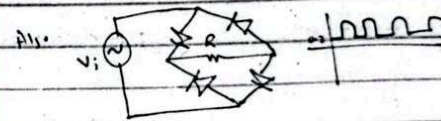


This circuit is good but very bulky and also expensive

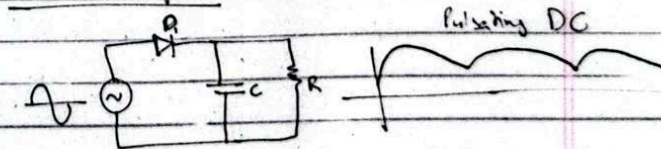


This circuit is very handy and also cost effective

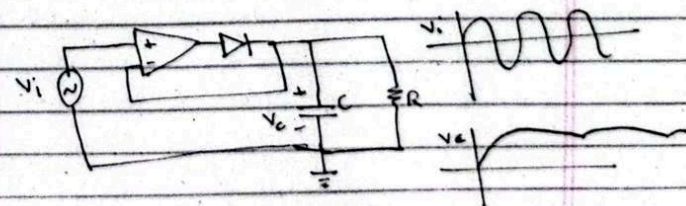
① & ② are both same.



Peak Rectifier



Precision Peak Rectifier



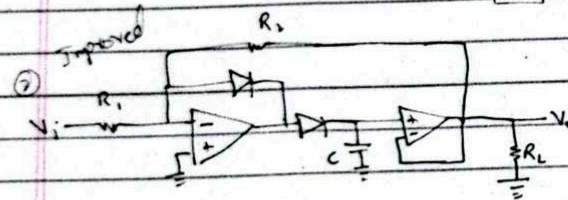
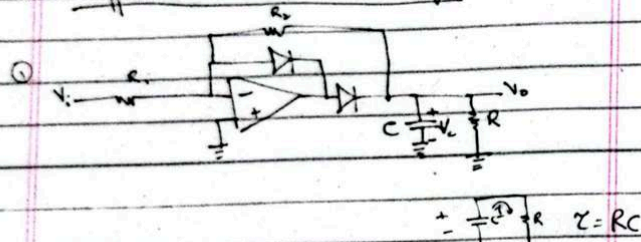
TTL: Transistor to Transistor Logic
ECL: Emitter Coupled Logic

Digital IC technologies & Logic CCTs Families

Logic cat. families are chosen as follows:-

- 1) Logic flexibility
- 2) Speed of operation
- 3) Availability of complex function
- 4) Noise immunity
- 5) Operating temp range
- 6) Power dissipation
- 7) Constant
- 8) Size

Buffered Precision Rectifier

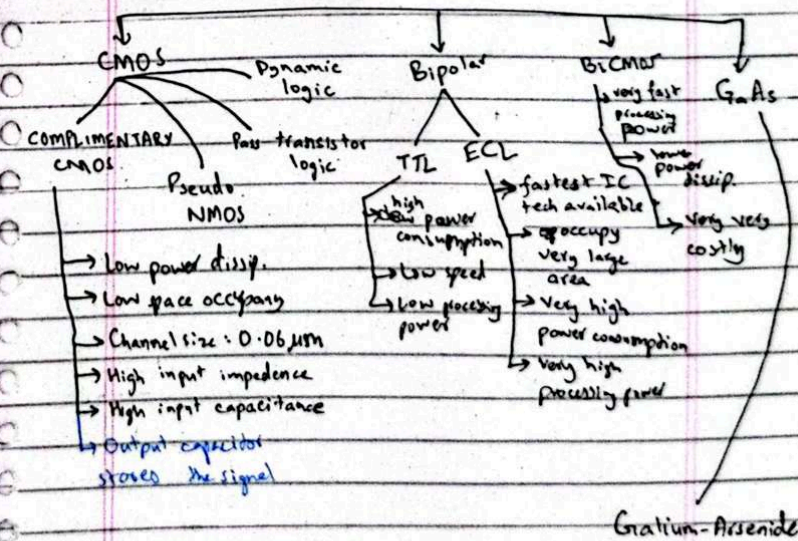
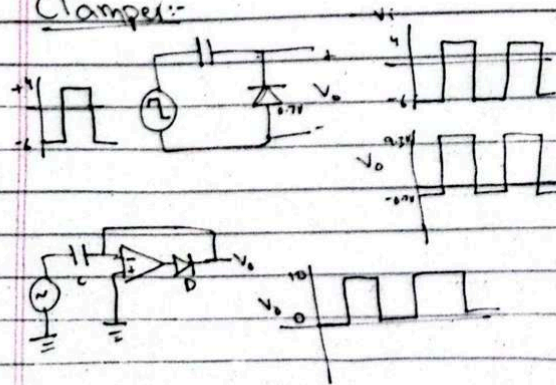


Pencil: Original Peak
 Red: Sidel

Black: ~~But~~ Precision Peak Rectifier

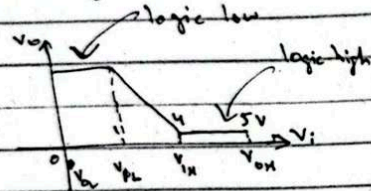
Blue: Buffered Precision Rectifier

Clamper:-



Logic ckt. Characteristics:

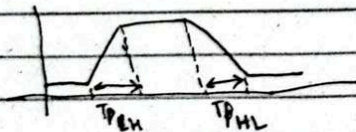
1) Noise Margin



$$NM_H = V_{OH} - V_{IH}$$

$$NM_L = V_{IL} - V_{OL}$$

2) Propagation Delay



3) Power Dissipation

1) Static dissip. $\Rightarrow P_S = \frac{V_{CC}^2}{R}$

2) Dynamic dissip. $\Rightarrow P_D = f C V_{CC}^2$

Delay-Power Product

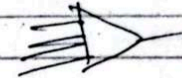
$$DP = TP \times PD$$

Less $TP \Rightarrow$ good IC Less $PD \Rightarrow$ good IC

Less $DP \Rightarrow$ Better IC

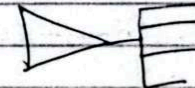
Fan In:

max no. of inputs



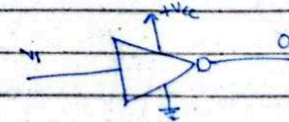
Fan Out

Max no. of loads



Digital Logic Inverter

1) Function of Inverter



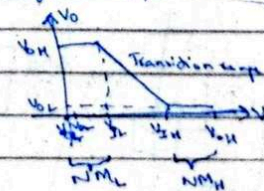
$$(0.1 \rightarrow 2.0V) \rightarrow 0$$

$$(3.5 \rightarrow 5.0) \rightarrow 1$$

$$'1' \rightarrow V_{OH} = 5V$$

$$'0' \rightarrow V_{OL} = 0.1V$$

2) Voltage-Transfer X-t-x



0-7
 n-type: 0 → open
 p-type: 1 → qm
 0 → close

→ Noise Margin

When $V_{OH} < V_i < V_{IL}$: $V_o = V_{OH}$

When $V_{IH} < V_i < V_{OH}$: $V_o = V_{OH}$

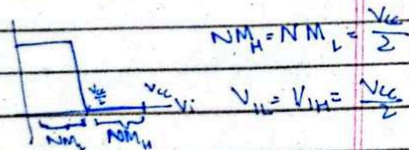
V_{IH} → max. input of an inverter that can be interpreted as logic low (0)

V_{IL} → min. input of an inverter that can be interpreted as logic high (1)

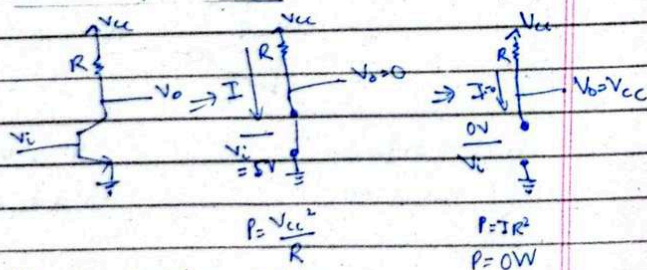
$$NM_L = V_{IL} - V_{OL}$$

$$NM_H = V_{OH} - V_{IH}$$

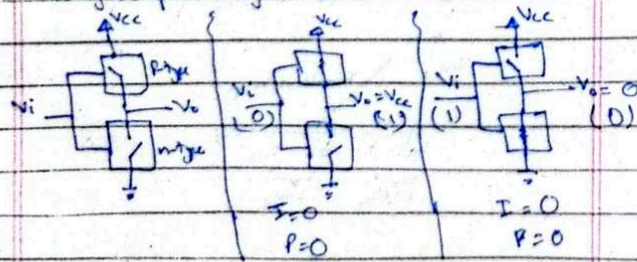
→ Ideal Inverter:



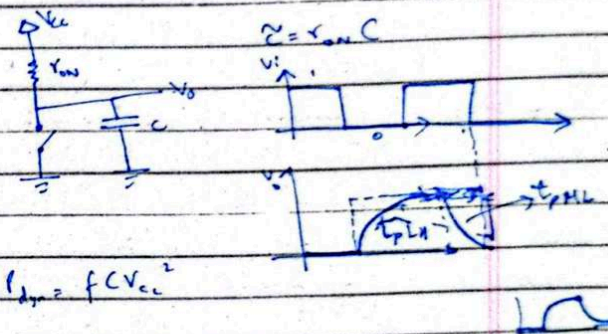
→ Inverter Implementation:



Using Complementary Switches:

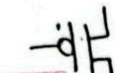


→ Dynamic Power Dissipation:-



NMOS 

PUN: Pull up network
PDN: Pull down network

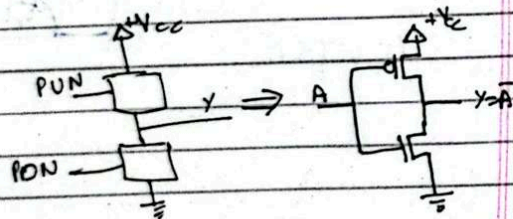
PMOS  $\rightarrow$  $\Rightarrow$ 

CMOS Logic Gate Circuits:-

Logic-level Inverter contains a
PUN & PDN

NMOS transistor $\rightarrow$ PDN

PMOS transistor $\rightarrow$ PUN



Operates in complementary function

$\rightarrow$ $A = '1'$, NMOS is ON, PMOS is OFF, $Y = 0$

$\rightarrow$ $A = '0'$, NMOS is OFF, PMOS is ON, $Y = 1$

For other logic gates

$\rightarrow$ PDN contains NMOS transistor

only — conducts when gate
input is high.

$\rightarrow$ PUN contains PMOS transistor

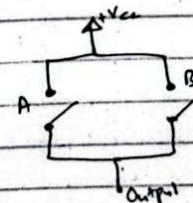
only — conducts when gate
input is low.

PUN & PDN are always complementary
to each other

TWO NMOS transistors in PDN
or TWO PMOS transistors
in PUN:-

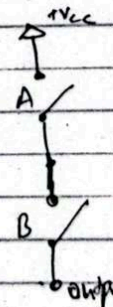
- 1) Perform OR function when
connected in parallel
- 2) Perform AND function when
connected in series.

OR



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

AND



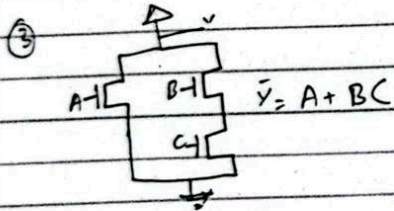
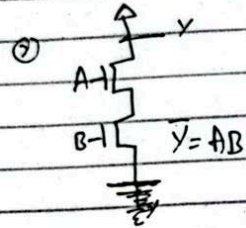
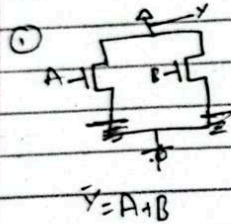
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

PUN is exact opposite of PDN.
i.e. two series NMOS transistors in
PDN are interpreted as two parallel
PMOS transistors in PUN.

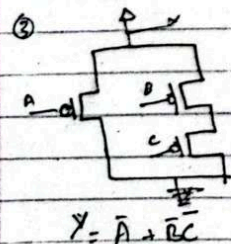
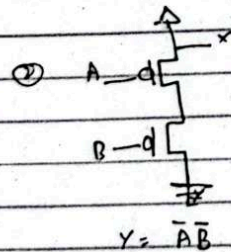
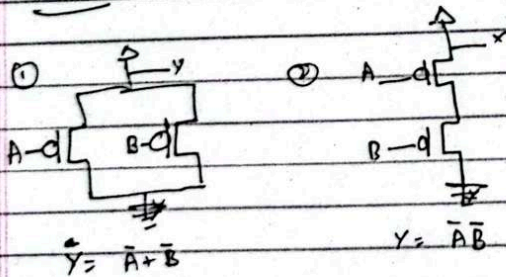
A

Gate V_{DD}
+

PDN



PUN



1st Method of CMOS Implementation

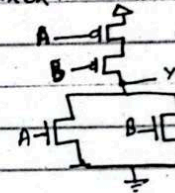
Two IP NOR Gate

$$Y = A + B$$

PUN: $Y = \overline{A+B}$

PDN: $\bar{Y} = \overline{A+B} = A+B$

Implement PDN first $\Rightarrow$ Complement PDN to get PUN

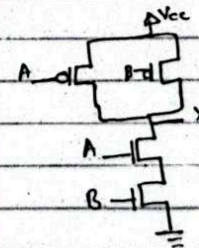


TWO IP NAND Gate

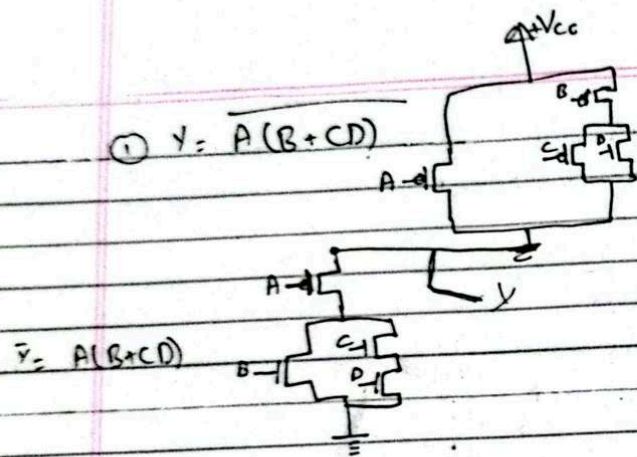
$$Y = AB$$

PUN: $Y = \overline{AB} = \bar{A} + \bar{B}$

PDN: $\bar{Y} = \overline{\bar{A} + \bar{B}} = AB$

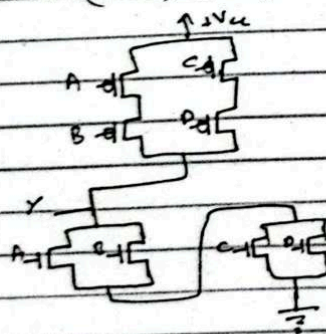


$$\textcircled{1} Y = \overline{A(B+CD)}$$



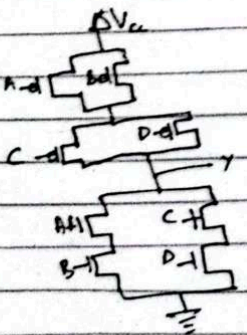
$$\textcircled{2} Y = \overline{(A+B)(C+D)}$$

$$\bar{Y} = (A+B)(C+D)$$



$$\textcircled{3} Y = \overline{AB+CD}$$

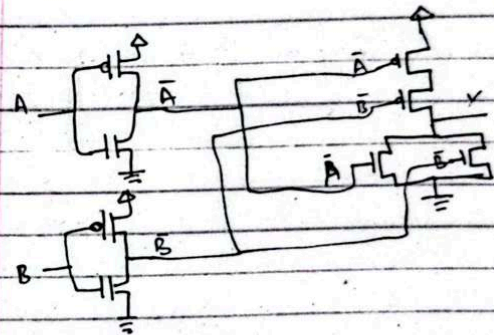
$$\bar{Y} = AB+CD$$



(Pre Inversion) 2nd Method of CMOS Implementation

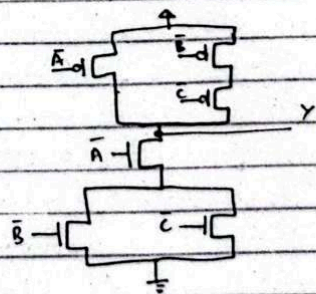
$$\textcircled{1} Y = AB$$

$$\bar{Y} = \overline{AB} = \bar{A} + \bar{B}$$



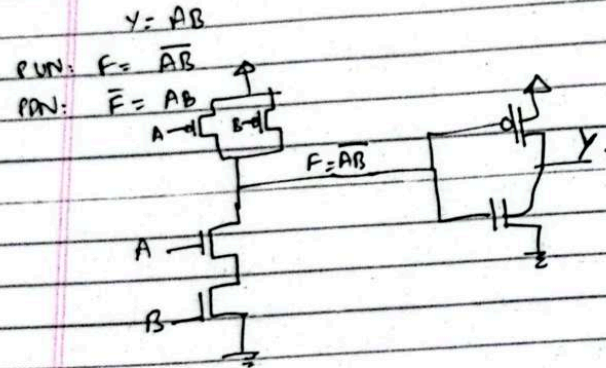
$$\textcircled{2} Y = A + \overline{BC}$$

$$\bar{Y} = \overline{A + \overline{BC}} = \bar{A}(\bar{B} + \bar{C})$$

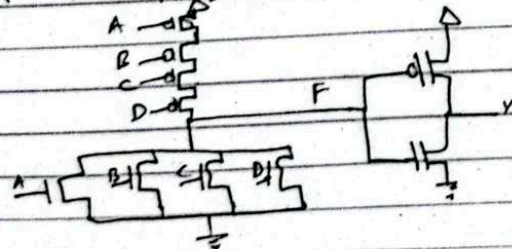


Post Inversion
3rd

Method of CMOS Implementation

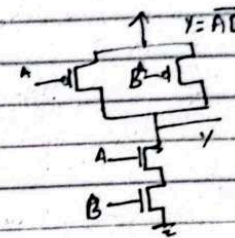


$\Rightarrow Y = A + B + C + D$
 PUN: $F = \overline{A + B + C + D}$
 PDN: $\overline{F} = A + B + C + D$

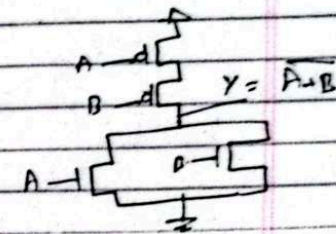


1st Method

NAND gate

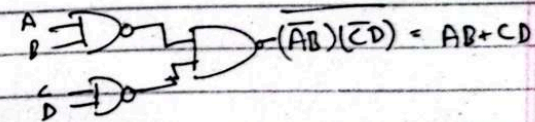


NOR Gate

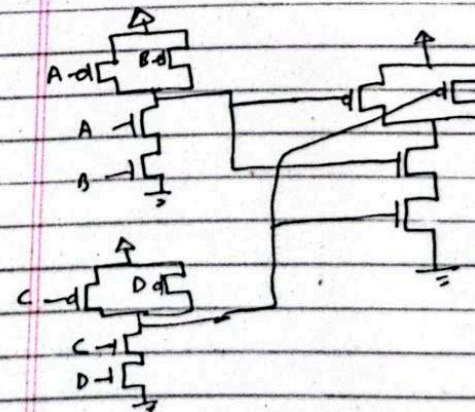


① $Y = AB + CD$

Make CMOS circuit using NAND gate



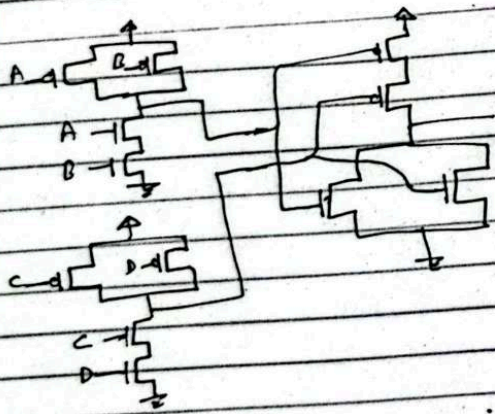
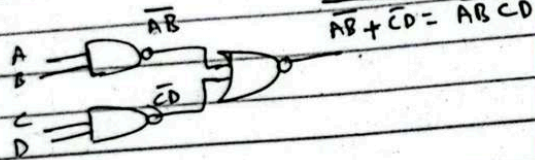
Place NAND circuit here



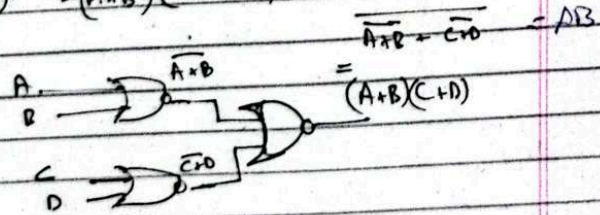
R4-870
R5-904

$\overline{AB+CD}$ $\overline{AB} \overline{CD}$ $\overline{A+B}$
 $\overline{A+B}$

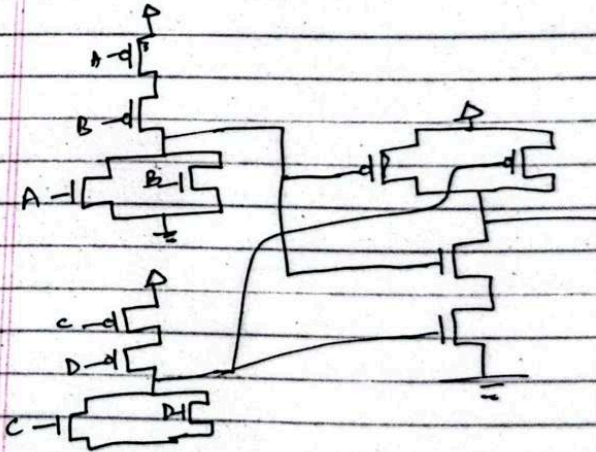
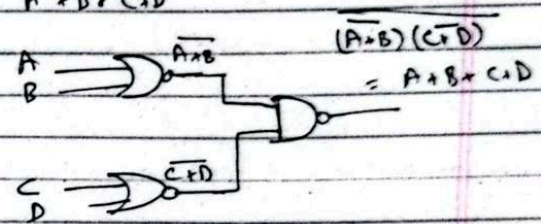
② $Y = ABCD$



③ $Y = (A+B)(C+D)$



④ $Y = A+B+C+D$



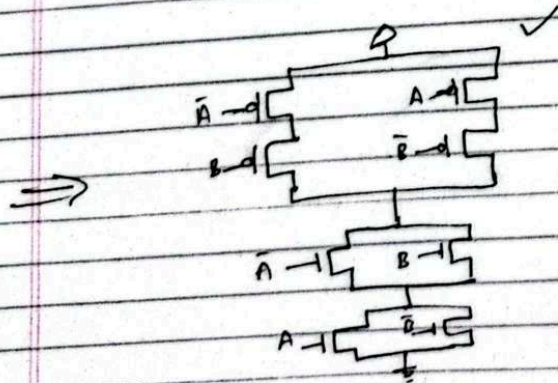
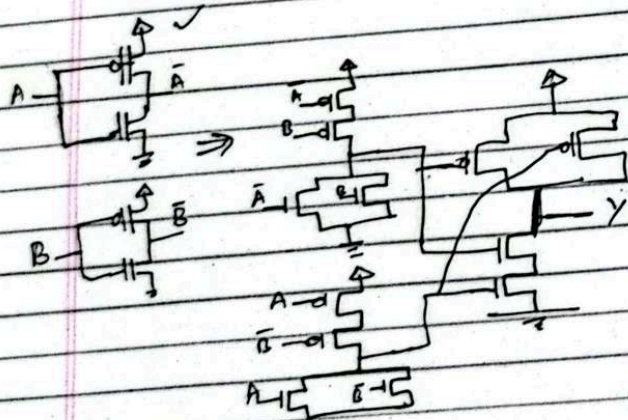
⑤ $Y = AB + C + D$

⑥

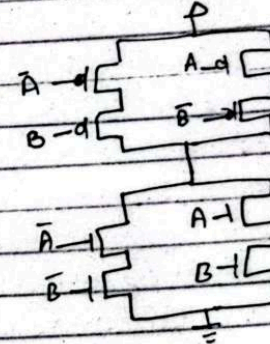
⑦ $Y = (A+B)AB(C+D)$

XOR Gate

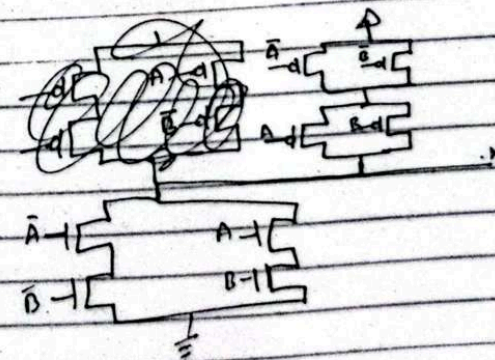
① PUN: $Y = A\bar{B} + \bar{A}B$
 PDN: $\bar{Y} = \overline{A\bar{B} + \bar{A}B} = (\bar{A} + B)(A + \bar{B})$



② PUN: $Y = A\bar{B} + \bar{A}B$
 PDN: $\bar{Y} = \overline{A\bar{B} + \bar{A}B} = (\bar{A} + B)(A + \bar{B})$
 $= \bar{A}A + \bar{A}\bar{B} + AB + B\bar{B}$
 $= \bar{A}\bar{B} + AB$



③ PUN: $Y = (A+B)(\bar{A}+\bar{B})$
 PDN: $\bar{Y} = \overline{(A+B)(\bar{A}+\bar{B})} = \overline{A+B} + \overline{\bar{A}+\bar{B}} = \bar{A}\bar{B} + AB$



Q. PUN: $Y = (A+B)(\bar{A}+\bar{B})$

PDN: $\bar{Y} = (\bar{A}+B)(A+\bar{B})$

